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(54) **EARPHONES**

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(57) **ABSTRACT**

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The present disclosure relates to an earphone (100), comprising a shell assembly (10) and an air-conducting loudspeaker (12). The shell assembly (10) is provided with an accommodation cavity (1001 and 1002) and a sound outlet hole (1080) for connecting the accommodation cavity (1001 and 1002) to an external environment. The air-conducting loudspeaker (12) includes a magnetic circuit assembly (120), a voice coil (122), and a vibration diaphragm (123). The magnetic circuit assembly (120) includes a shield (121) having an open end (1211), and an annular flange (1212) that is disposed at the open end (1211) of the shield (121) and protrudes from an outer periphery surface of the shield (121). A rim of the vibration diaphragm (123) is fixed to the annular flange (1212) and the vibration diaphragm (123) covers the open end (1211) of the shield (12). The voice coil (122) is connected to the vibration diaphragm (123), and the vibration diaphragm (123) is disposed on a side of the shield (121) away from the sound outlet hole (1080). Through the above design, the present disclosure can effectively improve the spatial utilization of the shell assembly (10), reduce the overall volume of the entire shell assembly (10), and also effectively enhance the sound quality.

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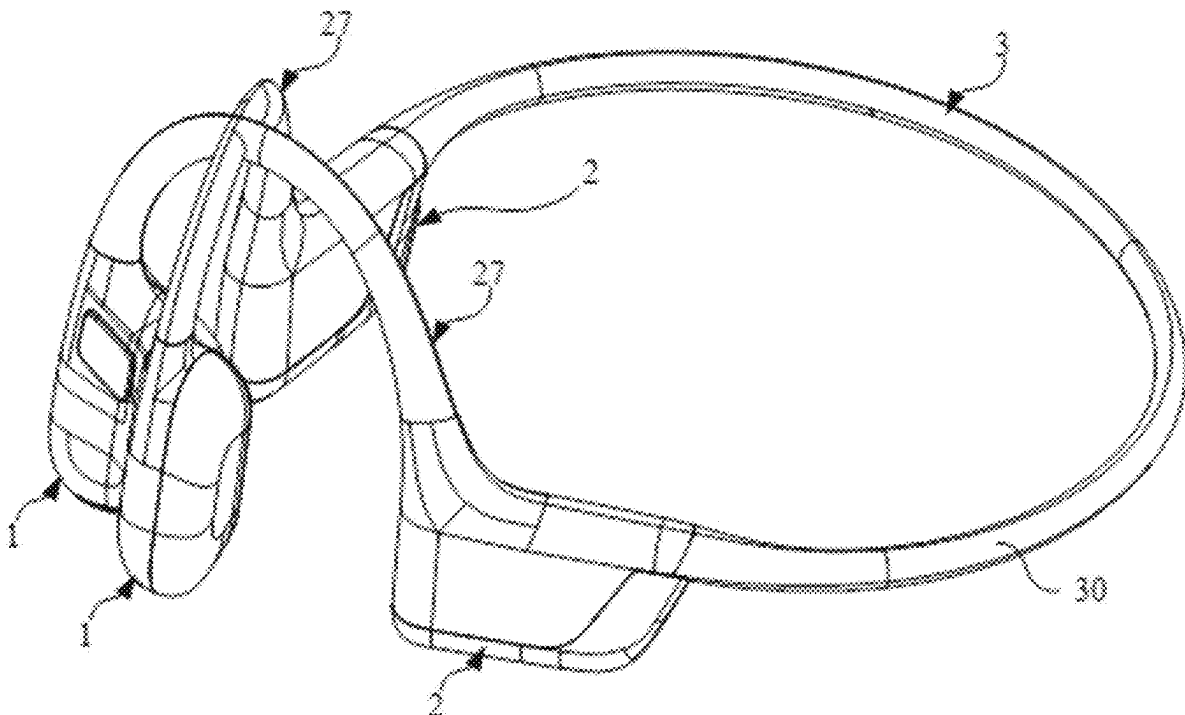
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H04R 1/10 (2006.01)



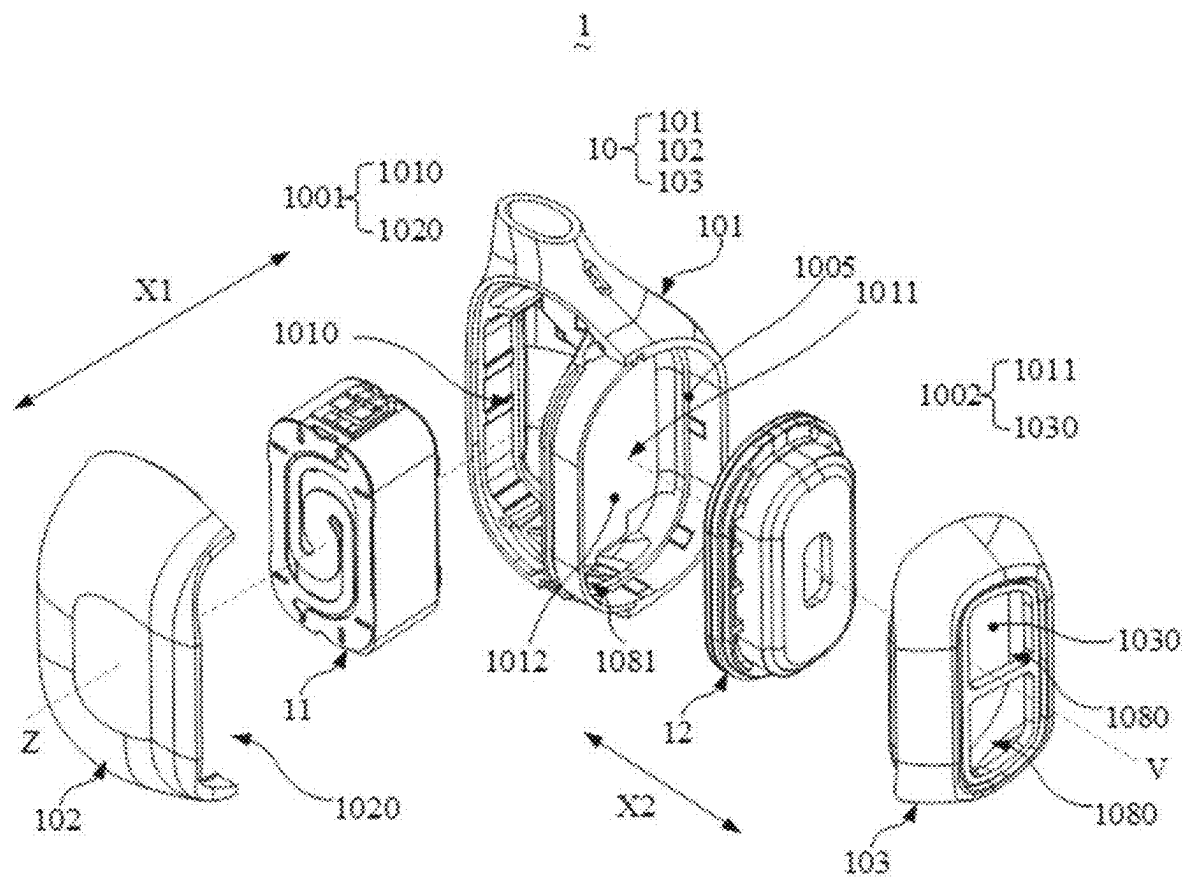


FIG. 3

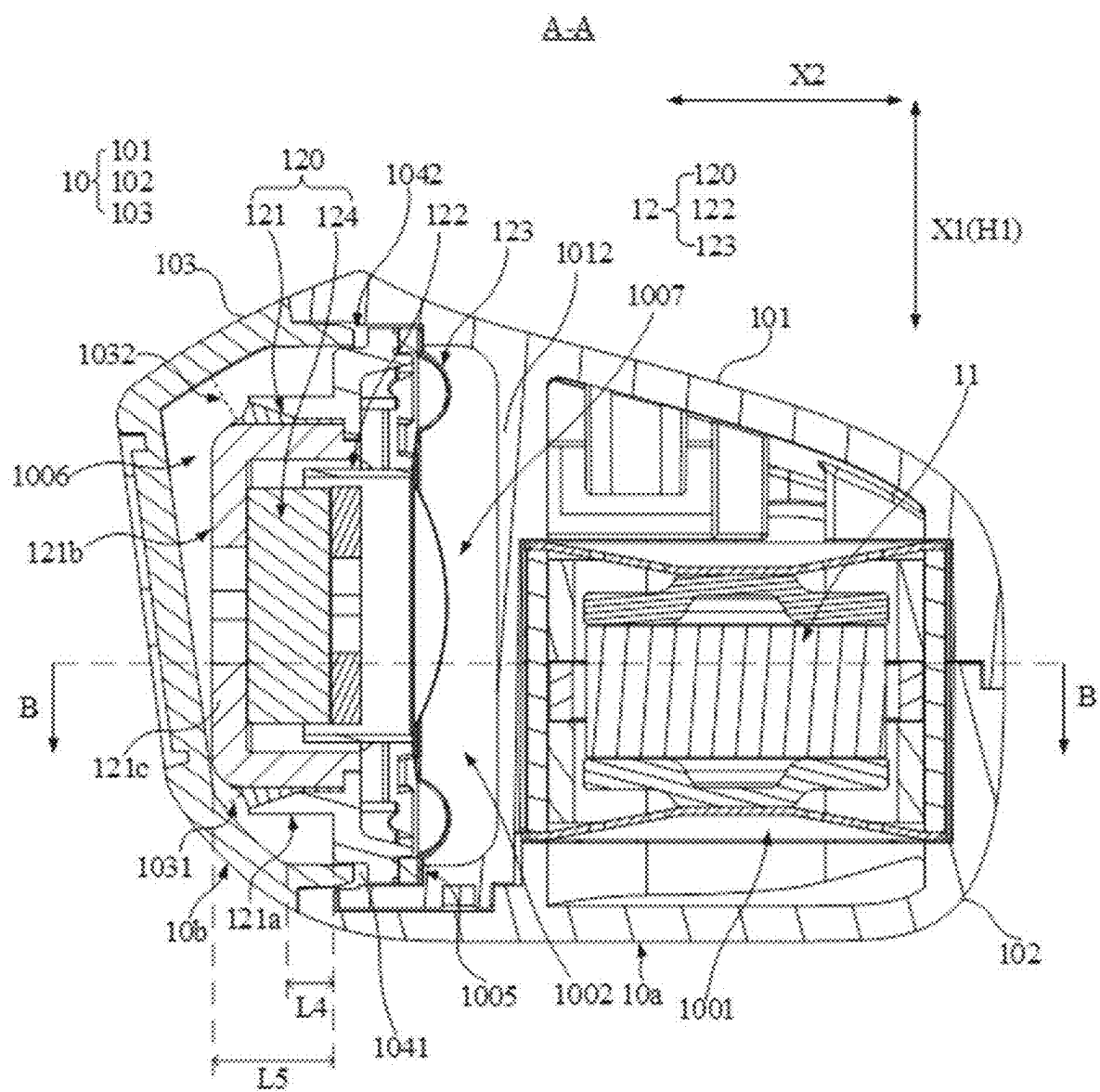


FIG. 4

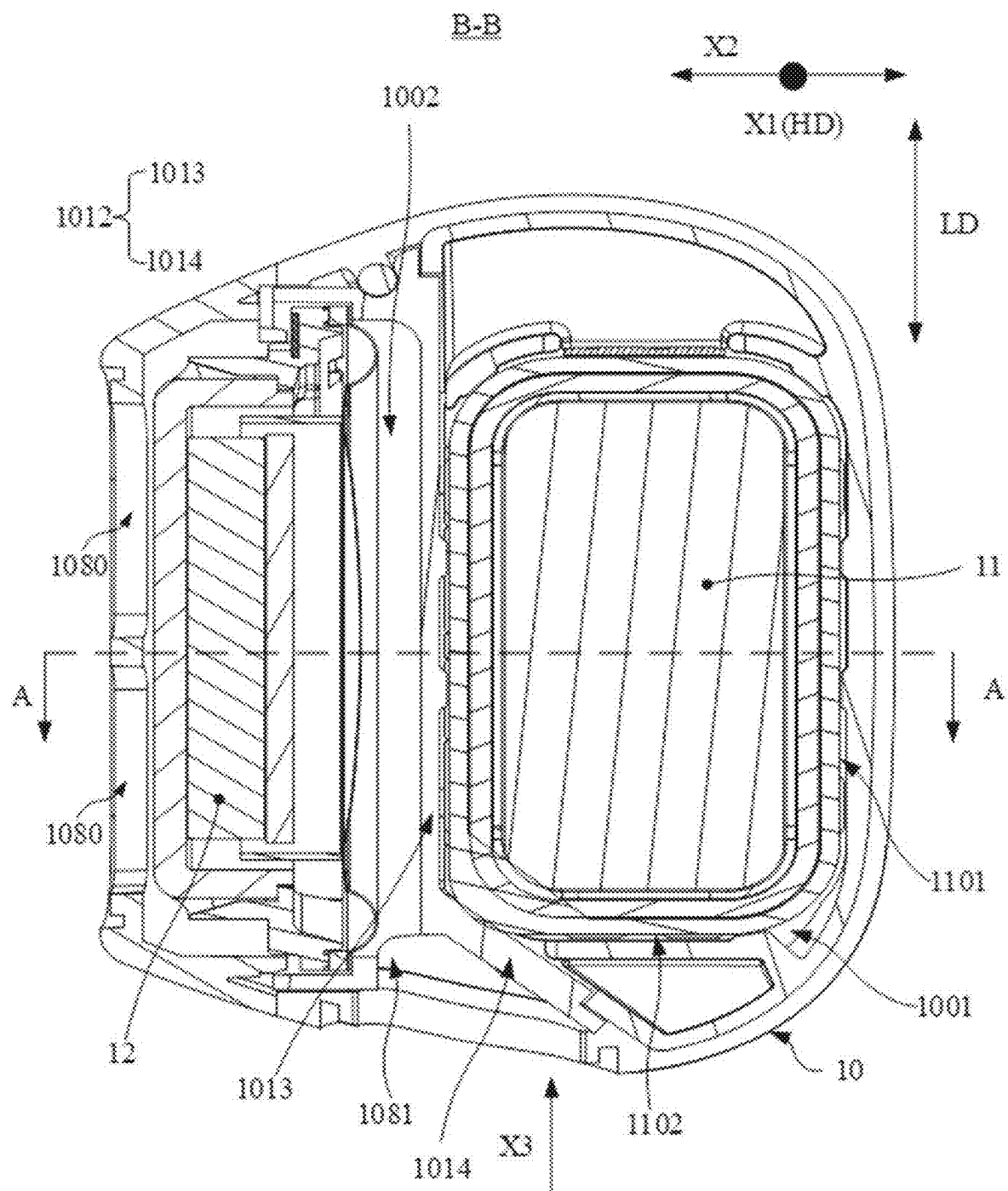


FIG. 5

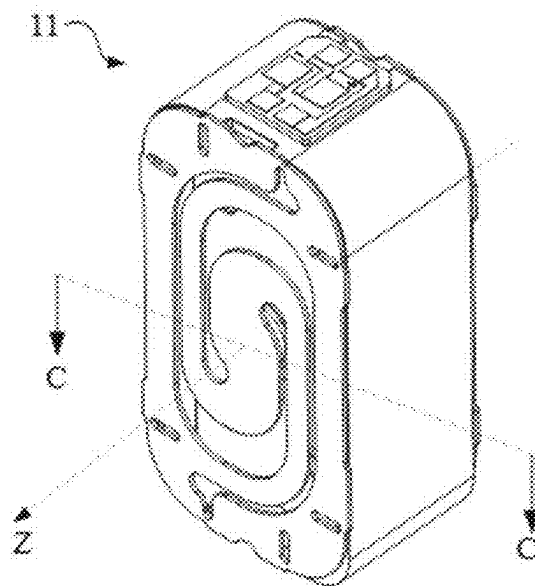


FIG. 6

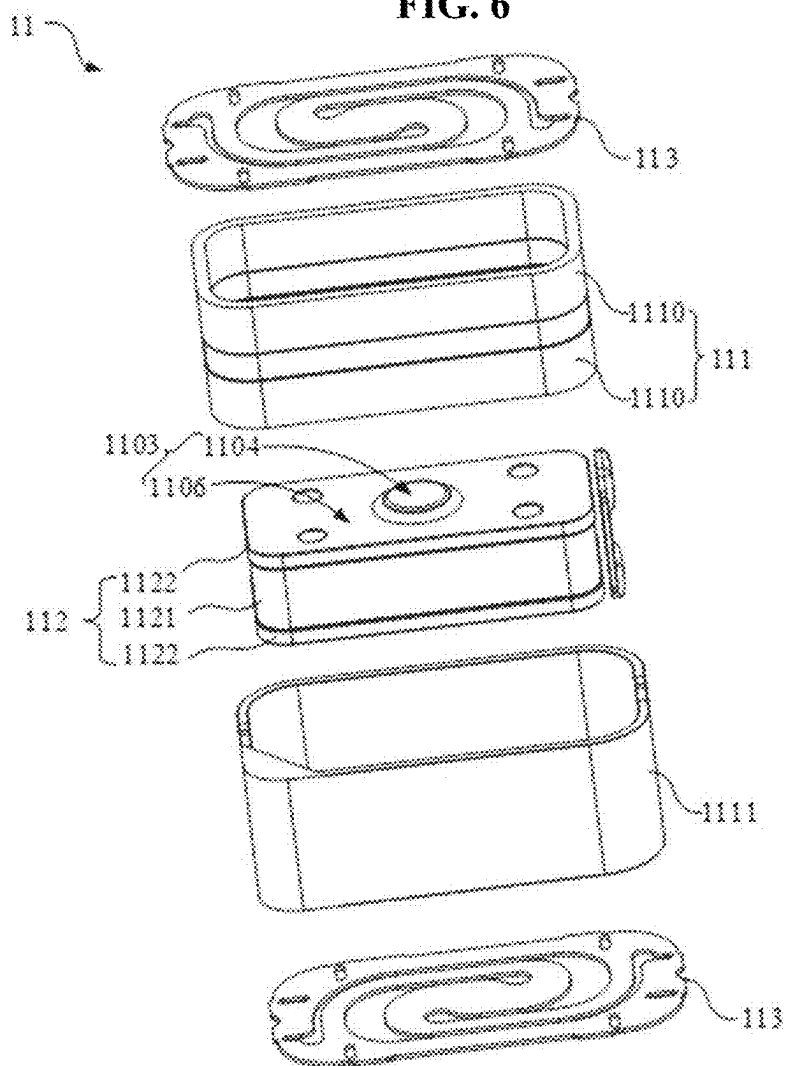


FIG. 7

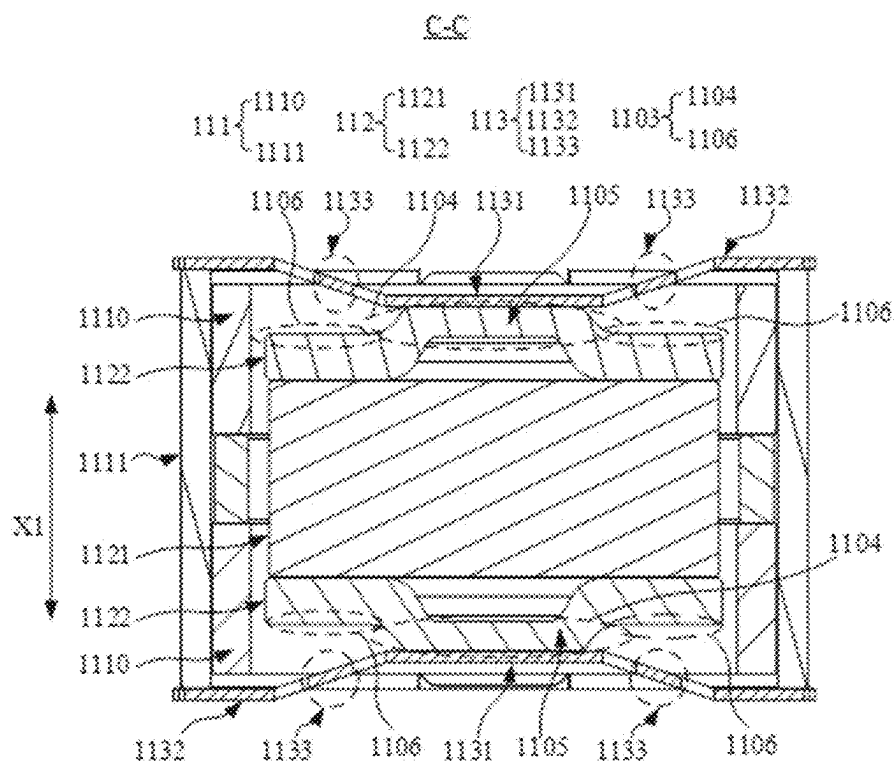


FIG. 8

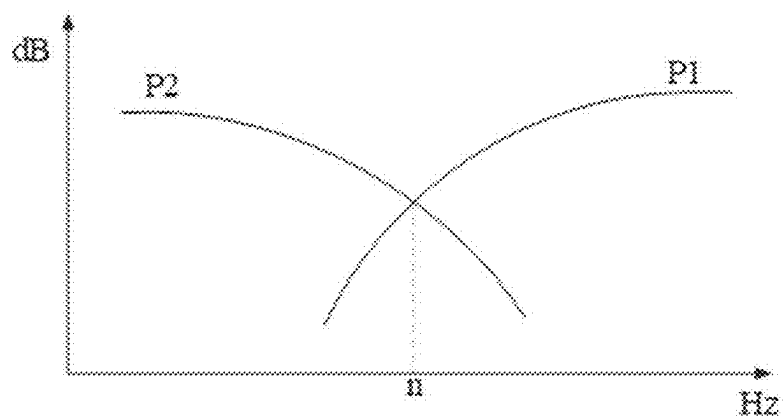


FIG. 9a

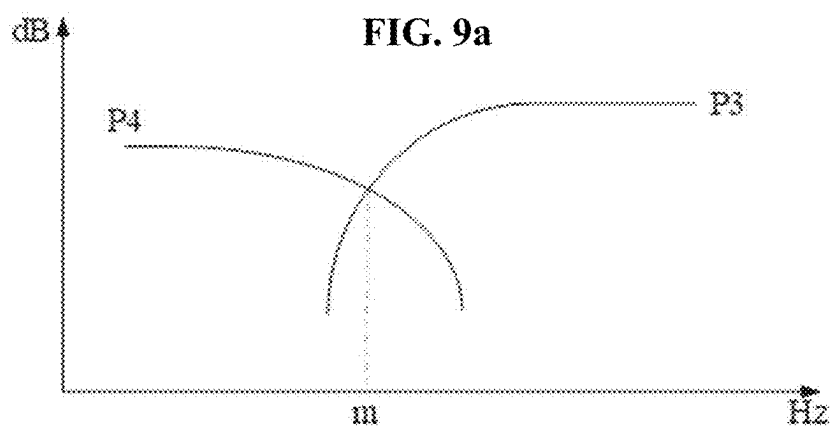
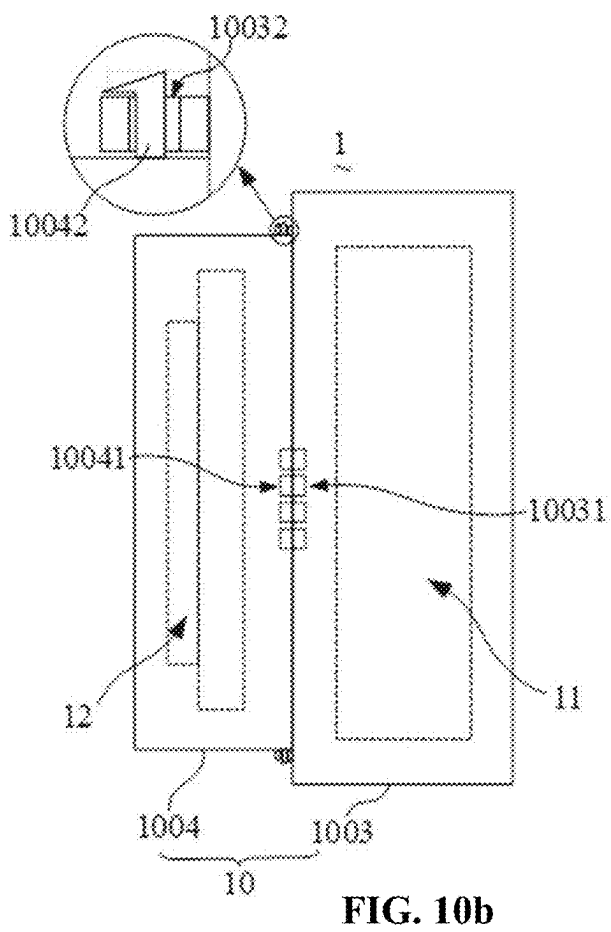
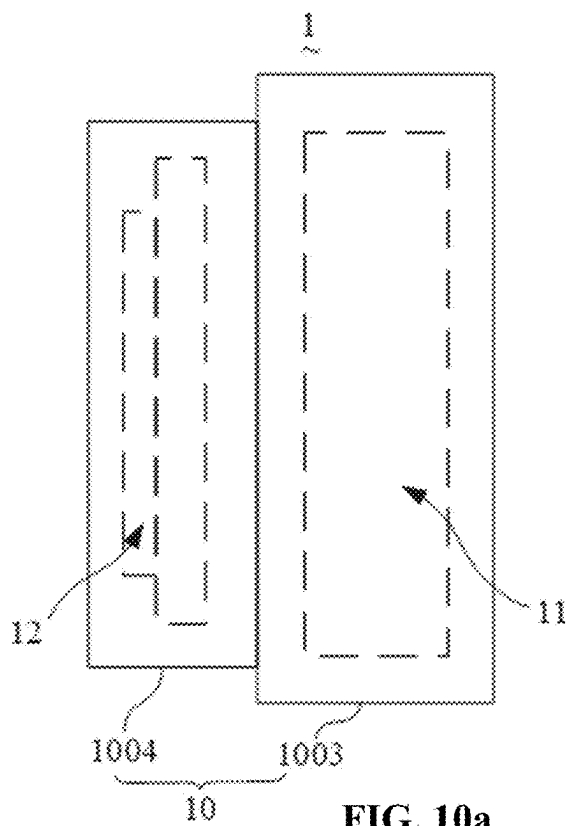


FIG. 9b



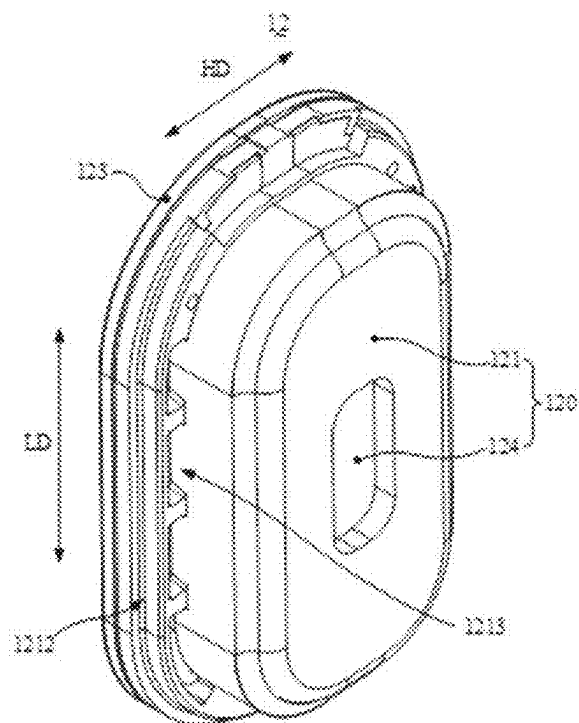


FIG. 11

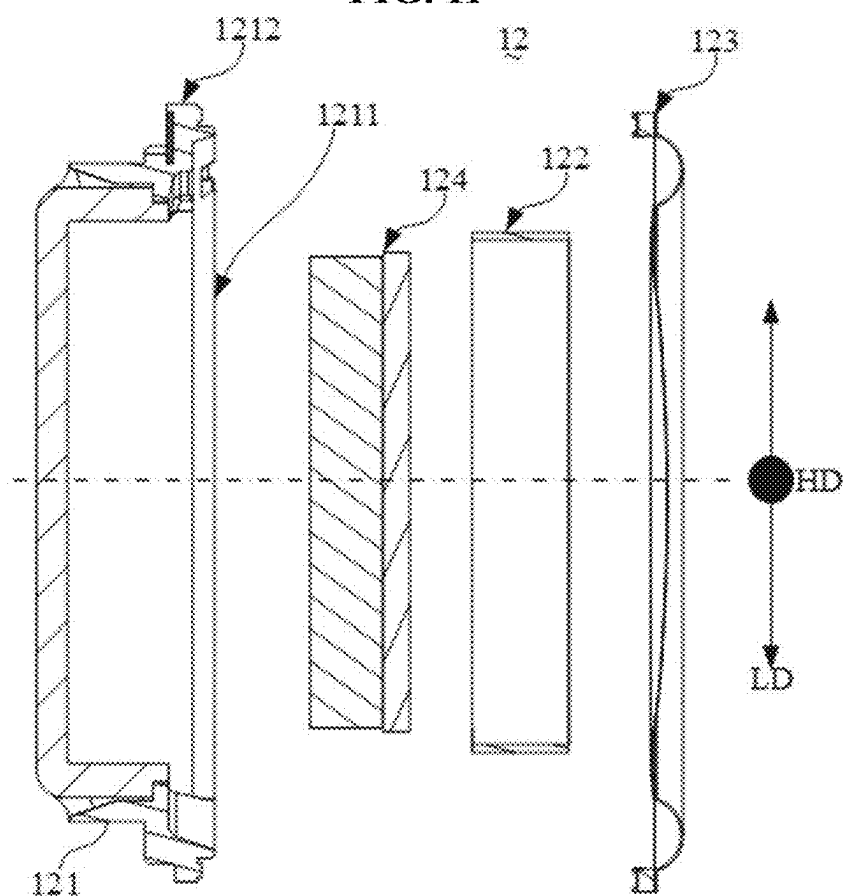


FIG. 12

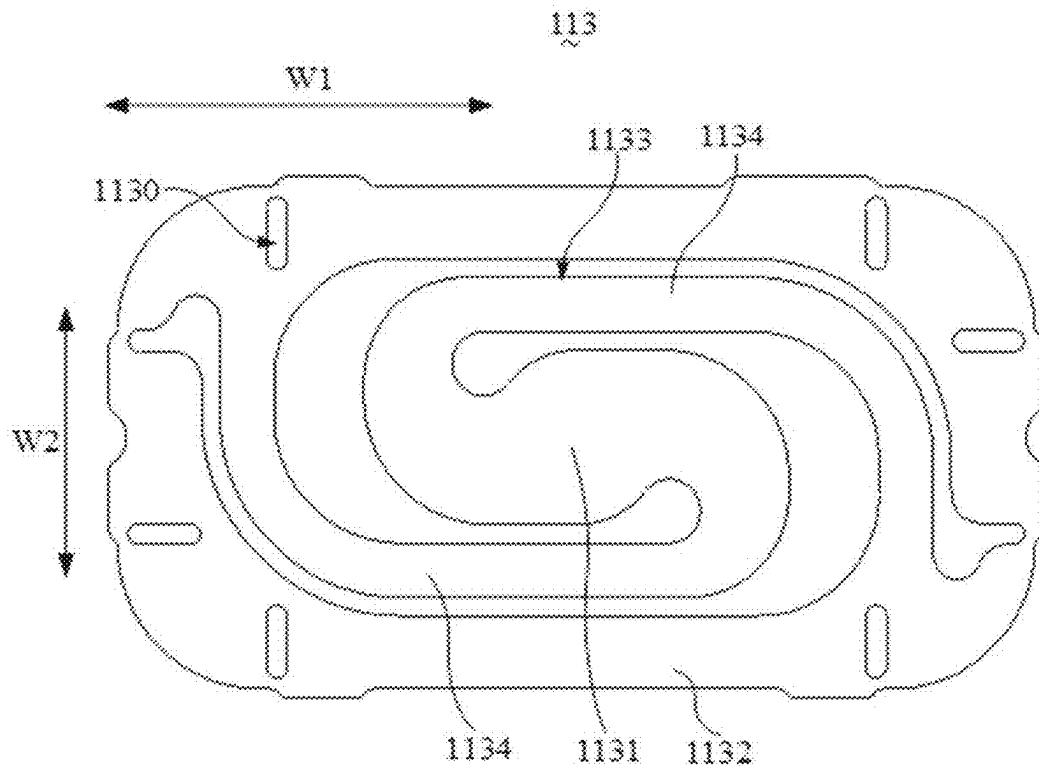


FIG. 13

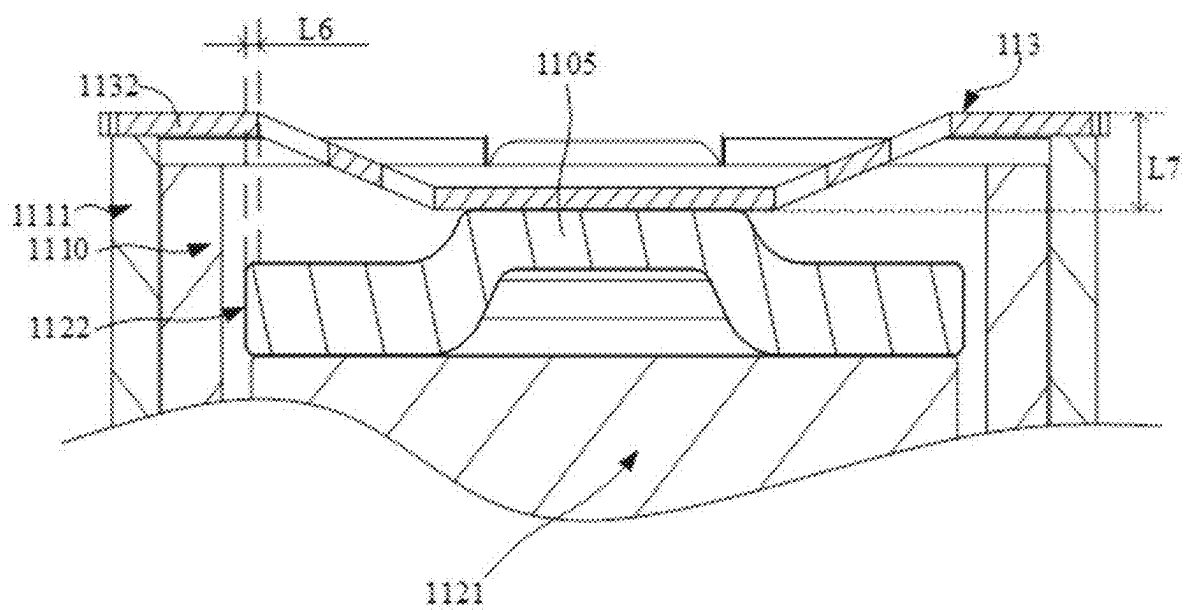


FIG. 14

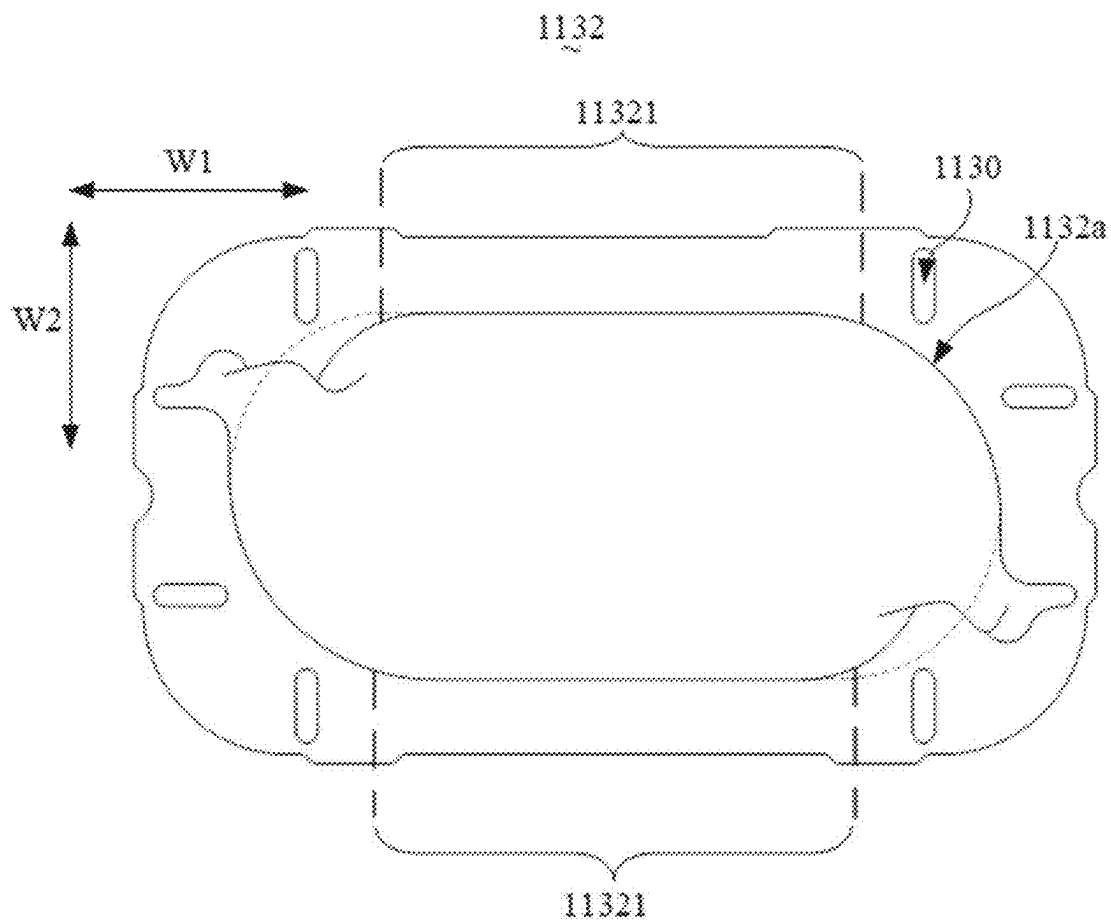


FIG. 15

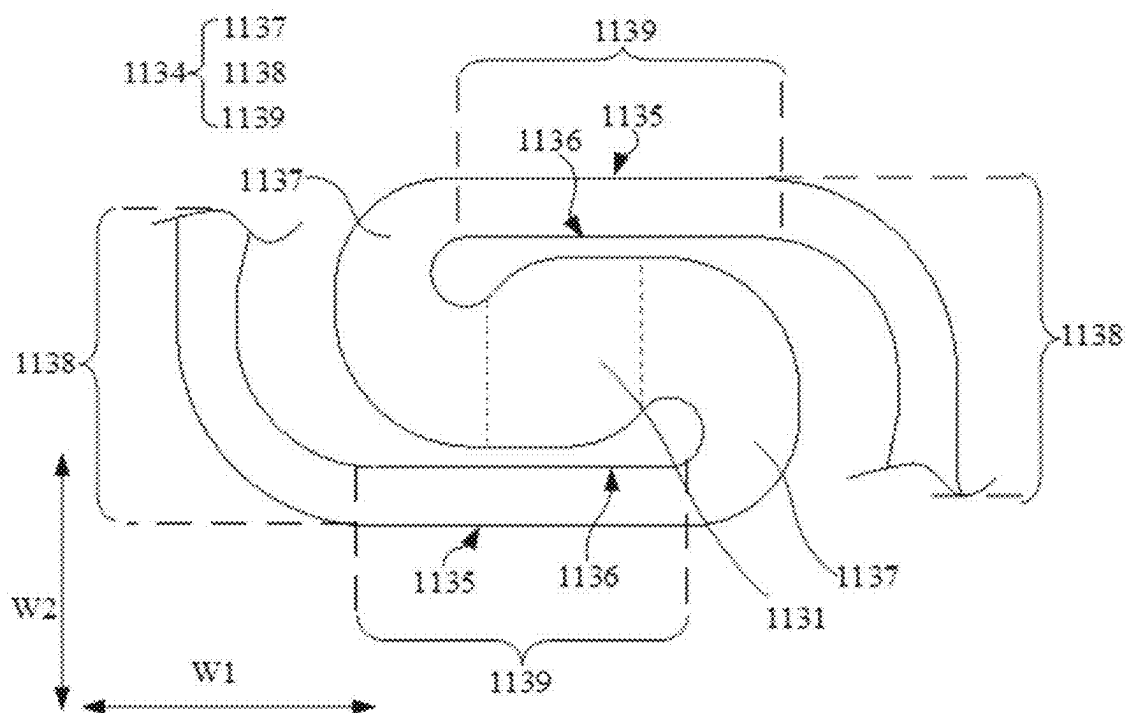


FIG. 16



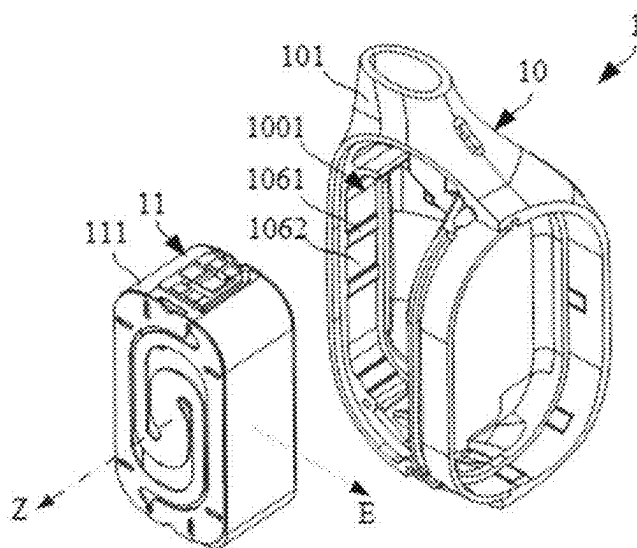


FIG. 19

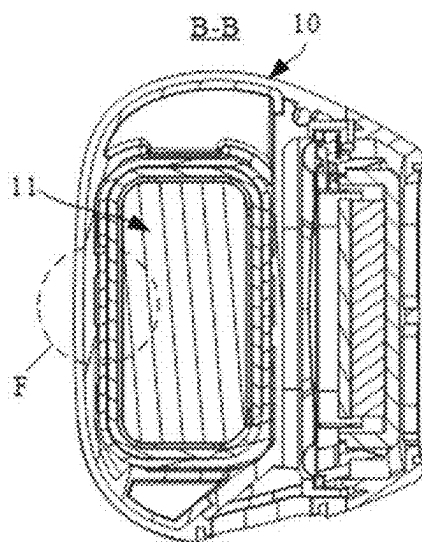


FIG. 20

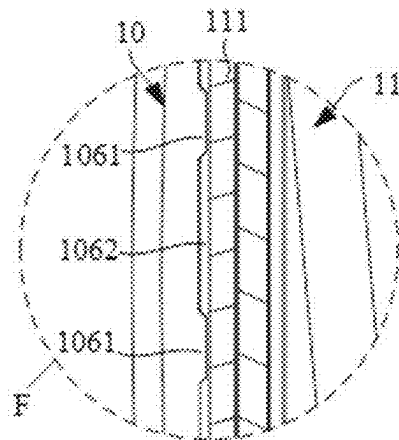


FIG. 21

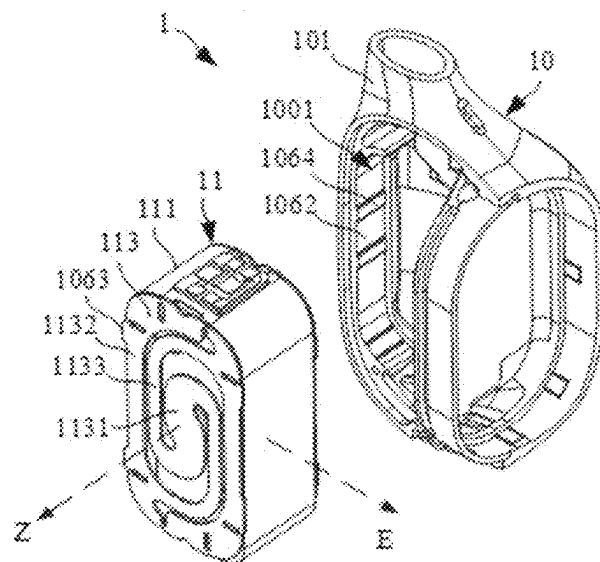


FIG. 22

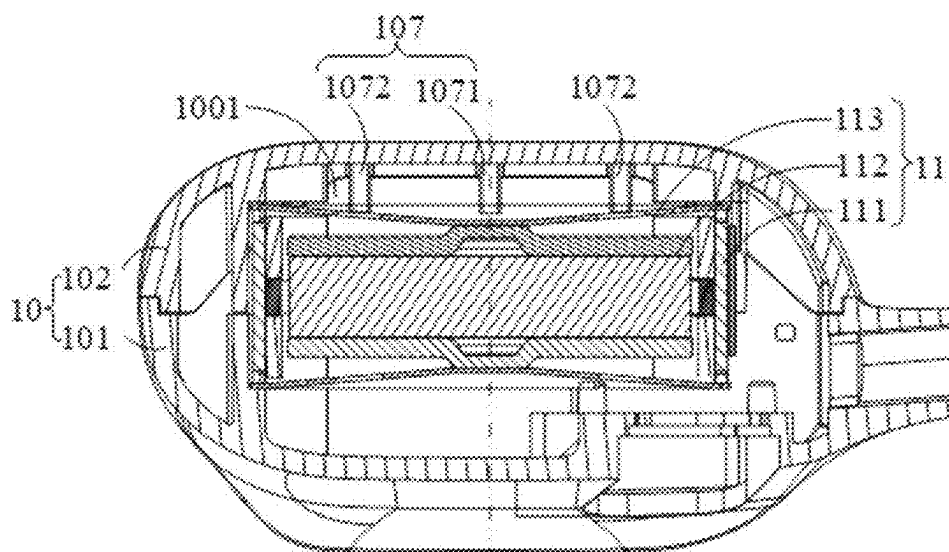


FIG. 23

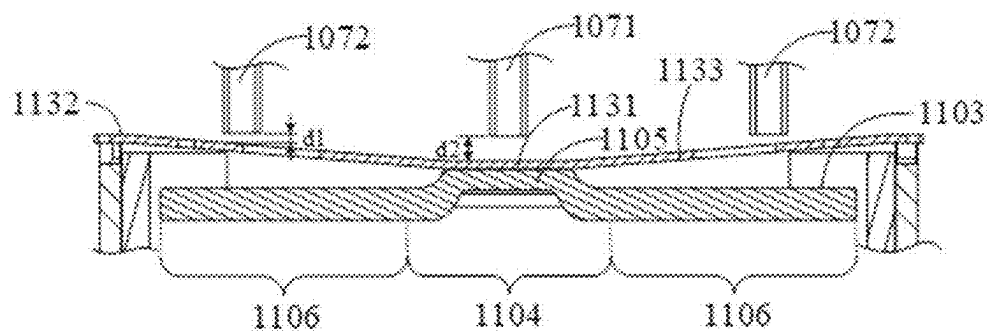


FIG. 24

EARPHONES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of International Application No. PCT/CN2023/140263, filed on Dec. 20, 2023, which claims priority to Chinese Patent Application No. 202310541798.X, entitled Earphones, filed on May 12, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present application relates to the technical field of electronic devices, and in particular, to earphones.

BACKGROUND

[0003] Earphones have been widely used in people's daily lives, as they can be paired with electronic devices such as smartphones and computers to provide users with an auditory feast. Currently, the principle of air-conducting sound transmission has been widely applied in earphones, and earphones using the air-conducting sound transmission are known for their high sound quality, making them popular among users. Typically, existing structures of air-conducting loudspeakers are relatively large in size. To optimize the sound path of the air-conducting loudspeaker, earphones often require a larger shell assembly. However, a larger shell assembly significantly increases the overall size of the earphones, which greatly affects the user's experience.

SUMMARY

[0004] The present disclosure provides an earphone, comprising: a shell assembly, the shell assembly being provided with accommodation cavities and a sound outlet hole for connecting the accommodation cavities to an external environment; and an air-conducting loudspeaker, the air-conducting loudspeaker including a magnetic circuit assembly, a voice coil, and a vibration diaphragm, the magnetic circuit assembly including a shield having an open end, and an annular flange being disposed at the open end of the shield and protruding from an outer periphery of the shield, a rim of the vibration diaphragm being fixed to the annular flange and the vibration diaphragm covering the open end of the shield, the voice coil being connected to the vibration diaphragm, and the vibration diaphragm being disposed on a side of the shield back away from the sound outlet hole.

[0005] In some embodiments, an annular table surface is provided on a wall of the accommodation cavities, the annular flange is pressed against the annular table surface, the vibration diaphragm separates the accommodation cavities into a first cavity portion and a second cavity portion, the first cavity portion is located on a side of the vibration diaphragm back away from the sound outlet hole, the second cavity portion is located on a side of the vibration diaphragm towards the sound outlet hole, the magnetic circuit assembly is located in the second cavity portion, and the shell assembly is provided with a pressure relief hole for connecting the first cavity portion to the external environment.

[0006] In some embodiments, the vibration diaphragm is located on a side of the annular flange towards the annular table surface, and the earphone further comprises a sealant,

and the sealant seals a gap between the shell assembly and a side of the annular flange back away from the annular table surface.

[0007] In some embodiments, the annular flange fits with the shell assembly to form an adhesive-accommodation groove for accommodating the sealant on the side of the annular flange back away from the annular table surface.

[0008] In some embodiments, the annular flange is provided with a sound-conducting hole, the sound-conducting hole is configured to connect a space enclosed by the vibration diaphragm and the magnetic circuit assembly with the second cavity portion located outside to the magnetic circuit assembly.

[0009] In some embodiments, the shell assembly includes a first shell, a second shell, and a third shell, the third shell fits with the first shell and/or the second shell along a vibration direction of the vibration diaphragm, the third shell is configured to accommodate an end of the shield back away from the vibration diaphragm, the sound outlet hole is disposed on the third shell; the shield includes an end surface and a circumferential side surface, the end surface is connected to the circumferential side surface, the end surface is disposed opposite to the sound outlet hole; there is a first gap between a connection region between the end surface and the circumferential side surface located on a first side and the third shell, there is a second gap between a connection region between the end surface and the circumferential side surface located on a second side and the third shell, the first gap being smaller than the second gap, and the first side and the second side being opposite to each other.

[0010] In some embodiments, in a wearing state, the first side is located on a side of the magnetic circuit assembly proximate to the face of a user.

[0011] In some embodiments, a portion of the end surface adjacent to the circumferential side surface located on the first side is in contact with and fixed to the third shell, or the connection region between the end surface and the circumferential side surface located on the first side is in contact with and fixed to the third shell.

[0012] In some embodiments, on a cross-section perpendicular to the vibration direction of the vibration diaphragm, the shield includes a length direction and a width direction, a dimension of the shield along the length direction is larger than a dimension of the shield along the width direction, and the first side and the second side are arranged opposite to each other along the width direction.

[0013] In some embodiments, on the first side, an average distance between the annular flange and the third shell along the vibration direction of the vibration diaphragm is greater than or equal to 0.5 mm, and on the second side, the second gap is greater than or equal to 0.5 mm.

[0014] In some embodiments, at least a portion of an outer surface of the third shell is disposed inclinedly towards the first shell and the second shell along a direction from the second side to the first side.

[0015] In some embodiments, the shield is a magnetic conduction shield, and the magnetic circuit assembly further includes a magnet disposed within the magnetic conduction shield, and the voice coil extends into a magnetic gap formed by the magnetic conduction shield and the magnet.

[0016] In some embodiments, a distance between the vibration diaphragm and a wall of the accommodation cavities along the vibration direction of the vibration diaphragm is greater than 0.8 mm.

[0017] In some embodiments, the shell assembly is provided with a first accommodation cavity and a second accommodation cavity isolated from each other; a sealing performance of the first accommodation cavity is greater than a sealing performance of the second accommodation cavity; the earphone further comprises a bone-conducting loudspeaker accommodated in the first accommodation cavity; and the air-conducting loudspeaker is accommodated in the second accommodation cavity.

[0018] In some embodiments, the shell assembly includes the first shell, the second shell, and the third shell, the first shell and the second shell fit with each other to form the first accommodation cavity, the first shell and/or the second shell further form a portion of the second accommodation cavity, and the third shell and the first shell and/or the second shell fit with each other to form another portion of the second accommodation cavity.

[0019] In some embodiments, the third shell is provided with the sound outlet hole connecting to the second accommodation cavity, and the first shell and/or the second shell are provided with the pressure relief hole connecting to the second accommodation cavity.

[0020] In some embodiments, the pressure relief hole is provided to extend inclinedly towards a side where the first accommodation cavity is located.

[0021] In some embodiments, a vibration direction of the bone-conducting loudspeaker intersects with a vibration direction of the air-conducting loudspeaker, and the first shell and the second shell fit with each other along the vibration direction of the bone-conducting loudspeaker, and the third shell and the first shell and/or the second shell fit with each other along the vibration direction of the air-conducting loudspeaker.

[0022] In some embodiments, the bone-conducting loudspeaker includes a voice coil assembly and a magnet assembly, the voice coil assembly is arranged around the magnet assembly and is fixedly connected to the first shell and/or second shell, and a surface of the first shell and/or the second shell arranged opposite to the bone-conducting loudspeaker along the vibration direction of the bone-conducting loudspeaker is configured to contact the skin of the user and transmit bone-conducting vibration.

[0023] In some embodiments, the voice coil assembly includes two sets of voice coils spaced apart along a central axis direction of the bone-conducting loudspeaker and a magnetic conduction shield arranged around an outer periphery of the two sets of voice coils, the magnet assembly includes a magnet and two magnetic conduction plates, the two magnetic conduction plates are arranged on two opposite end surfaces of the magnet along the central axis direction of the bone-conducting loudspeaker, projections of the two magnetic conduction plates along a radial direction of the bone-conducting loudspeaker overlap with the two sets of voice coils, and current directions of the two sets of voice coils are opposite to each other.

[0024] In some embodiments, the earphone is provided with a first working mode and a second working mode, in the first working mode, the bone-conducting loudspeaker plays a sound signal of a first frequency band and the air-conducting loudspeaker plays a sound signal of a second frequency band; and in the second working mode, the bone-conducting loudspeaker plays the sound signal of the first frequency band and the sound signal of the second

frequency band, and the first frequency band and the second frequency band being at least partially staggered in a frequency domain.

[0025] In some embodiments, the earphone is further provided with a third working mode, and in the third working mode, the air-conducting loudspeaker plays the sound signal of the first frequency band and the sound signal of the second frequency band.

[0026] In some embodiments, the earphone comprises a wearing assembly, the shell assembly includes a first portion and a second portion, the first portion is connected to the wearing assembly, the second portion is connected to the first portion or the wearing assembly detachably, the bone-conducting loudspeaker is disposed on the first portion, and the air-conducting loudspeaker is disposed on the second portion and is detachable from the wearing assembly with the second portion.

[0027] In some embodiments, the first portion is provided with a first electrode portion, the second portion is provided with a second electrode portion, and when the first portion and the second portion are connected, the first electrode portion and the second electrode portion are in contact with each other and electrically connected.

[0028] In some embodiments, the first portion is provided with a first connection portion spaced apart from the first connection portion, the second portion is provided with a second connection portion, and the first connection portion and the second connection portion are provided to be connected detachably by at least one of snap-fitting, bonding, magnetic suction, or threaded connection.

[0029] In some embodiments, the shell assembly is provided with a first accommodation cavity and a second accommodation cavity isolated from each other; the earphone further comprises a bone-conducting loudspeaker accommodated in the first accommodation cavity, and the air-conducting loudspeaker is accommodated in the second accommodation cavity; the shell assembly is provided with the pressure relief hole for connecting the second accommodation cavity to the external environment; and the pressure relief hole is provided to extend towards a side where the first accommodation cavity is located.

[0030] In some embodiments, a projection of the pressure relief hole along a direction perpendicular to an arrangement direction of the first accommodation cavity and the second accommodation cavity and the first accommodation cavity at least partially overlap.

[0031] In some embodiments, the pressure relief hole is provided to extend inclinedly towards the side where the first accommodation cavity is located, and along a direction perpendicular to an arrangement direction of the first accommodation cavity and the second accommodation cavity, the pressure relief hole gradually increases in dimension along a direction from the second accommodation cavity to the external environment.

[0032] In some embodiments, the shell assembly includes a partition wall for isolating the first accommodation cavity and the second accommodation cavity, at least a portion of the partition wall extends toward the side where the first accommodation cavity is located and forms a portion of a wall of the pressure relief hole.

[0033] In some embodiments, on a cross-section perpendicular to a vibration direction of the bone-conducting loudspeaker, the bone-conducting loudspeaker includes a long-side edge and a short-side edge, a dimension of the

long-side edge is larger than a dimension of the short-side edge, the partition wall includes a first wall segment and a second wall segment, the first wall segment and the long-side edge are arranged side by side, the second wall segment is connected to the first wall segment and extends inclinedly from a periphery of a junction between the long-side edge and the short-side edge towards the side where the first accommodation cavity is located, and the second wall segment is used to form the portion of the wall of the pressure relief hole.

[0034] In some embodiments, the shell assembly includes the first shell, the second shell, and the third shell, the partition wall is disposed on the first shell and/or the second shell, the first shell and the second shell fit with each other to form the first accommodation cavity and the pressure relief hole, the first shell and/or the second shell further form a portion of the second accommodation cavity, and the third shell and with the first shell and/or the second shell fit with each other to form another portion of the second accommodation cavity.

[0035] In some embodiments, a vibration direction of the bone-conducting loudspeaker intersects with a vibration direction of the air-conducting loudspeaker, and the first shell and the second shell fit with each other along the vibration direction of the bone-conducting loudspeaker, and the third shell and the first shell and/or the second shell fit with each other along the vibration direction of vibration of the air-conducting loudspeaker.

[0036] In some embodiments, the partition wall is disposed on the first shell, the first shell is provided with a first sub-accommodation cavity and a second sub-accommodation cavity arranged on opposite sides of the partition wall, an opening direction of the first sub-accommodation cavity is disposed along a wall surface of the partition wall, an opening direction of the second sub-accommodation cavity intersects with the wall surface of the partition wall, the second shell is provided with a third sub-accommodation cavity, the second shell is capped at an open end of the first sub-accommodation cavity, the third sub-accommodation cavity and the first sub-accommodation cavity fit to form the first accommodation cavity, the third shell is provided with a fourth sub-accommodation cavity, the third shell is capped at an open end of the second sub-accommodation cavity, and the fourth sub-accommodation cavity fit with the second sub-accommodation cavity to form the second accommodation cavity.

[0037] In some embodiments, the air-conducting loudspeaker includes a vibration diaphragm, the vibration diaphragm separates the second accommodation cavity into a first cavity portion disposed on a side of the vibration diaphragm back away from the partition wall and a second cavity portion disposed between the vibration diaphragm and the partition wall, the pressure relief hole is connected to the second cavity portion, and the shell assembly is further provided with a sound outlet hole connected to the first cavity portion.

[0038] In some embodiments, the air-conducting loudspeaker further includes the magnetic circuit assembly and the voice coil, the magnetic circuit assembly includes the shield having the open end and the annular flange disposed at the open end of the shield and protruding from the outer periphery of the shield, the rim of the vibration diaphragm is fixed to a side of the annular flange towards the open end and the vibration diaphragm covers the open end of the

shield, the voice coil is connected to the vibration diaphragm, and the vibration diaphragm is disposed away from the sound outlet hole compared to the shield.

[0039] In some embodiments, a center of the pressure relief hole is not higher than a center of the sound outlet hole, and a line connecting the center of the pressure relief hole and the center of the sound outlet hole points towards the user's ear in a wearing state; and/or, in a wearing state, an angle between the line connecting the center of the pressure relief hole and the center of the sound outlet hole and a coronal axis of the user is $\geq 0^\circ$ and $< 90^\circ$, and an angle between the line connecting the center of the pressure relief hole and the center of the sound outlet hole and a vertical axis of the user $> 0^\circ$ and $\leq 90^\circ$.

[0040] In some embodiments, a central axis of the bone-conducting loudspeaker and a central axis of the air-conducting loudspeaker are skew lines, and along a direction perpendicular to the central axis of the bone-conducting loudspeaker and the central axis of the air-conducting loudspeaker, the central axis of the bone-conducting speaker is located farther from the pressure relief hole compared to the central axis of the air-conducting loudspeaker.

[0041] In some embodiments, a distance between the central axis of the bone-conducting loudspeaker and the central axis of the air-conducting loudspeaker is in a range of 0.2 mm to 1.2 mm.

[0042] The beneficial effects of the present disclosure are as follows: by arranging the annular flange and the vibration diaphragm on the side away from the sound outlet hole, the vibration diaphragm and the annular flange can be located closer to the interior of the shell assembly. With such an arrangement, in the air-conducting loudspeaker, structures with a larger radial dimension are arranged away from the sound outlet hole, and structures with a smaller dimension can be located closer to the sound outlet hole. Such an opposite configuration inside the air-conducting loudspeaker, compared to the traditional loudspeaker design where the vibration diaphragm must be oriented toward the sound outlet hole, effectively reduces the dimension of the portion of the shell assembly that is near the sound outlet hole, which allows the dimension of the shell assembly to decrease from the central region to the region around the sound outlet hole (i.e., the radial dimension of the outer peripheral surface of the shell assembly can gradually decrease). As a result, the structure of the shell assembly becomes more compact, significantly improving the spatial utilization and reducing the overall volume of the shell assembly. Moreover, such an opposite configuration can optimize the sound path and thus improve the sound quality.

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] To more clearly illustrate the technical solutions in the embodiments of the present disclosure, the accompanying drawings that need to be used in the description of the embodiments will be briefly introduced in the following, and it will be obvious that the accompanying drawings in the following description are only some of the embodiments of the present disclosure, and other attachments can be obtained according to them without creative labor to a person of ordinary skill in the art.

[0044] FIG. 1 is a schematic diagram illustrating a 3D structure of a general assembly of an earphone according to some embodiments of the present disclosure;

[0045] FIG. 2 is a schematic diagram illustrating a 3D structure of a portion between a core assembly and a wearing assembly shown in FIG. 1;

[0046] FIG. 3 is an exploded structural diagram of the core assembly shown in FIG. 2;

[0047] FIG. 4 is a schematic diagram illustrating a sectional structure of the core assembly shown in FIG. 2 along A-A;

[0048] FIG. 5 is a schematic diagram illustrating a sectional structure of the core assembly shown in FIG. 2 along B-B;

[0049] FIG. 6 is a schematic diagram illustrating a 3D structure of a bone-conducting loudspeaker shown in FIG. 3;

[0050] FIG. 7 is an exploded schematic diagram of a bone-conducting loudspeaker shown in FIG. 3;

[0051] FIG. 8 is a schematic diagram illustrating a sectional structure of the bone-conducting loudspeaker shown in FIG. 6 along C-C;

[0052] FIG. 9a is a schematic diagram illustrating sound signals in a first frequency band and a second frequency band in the earphone shown in FIG. 1;

[0053] FIG. 9b is a schematic diagram illustrating frequencies of electrical signals input to a bone-conducting loudspeaker and an air-conducting loudspeaker of the earphone shown in FIG. 1 in a first working mode;

[0054] FIG. 10a is a schematic diagram illustrating an exemplary structure of a detachable solution for a bone-conducting loudspeaker and an air-conducting loudspeaker of the earphone shown in FIG. 1;

[0055] FIG. 10b is a schematic diagram illustrating an exemplary structure of the bone-conducting loudspeaker and the air-conducting loudspeaker shown in FIG. 10a;

[0056] FIG. 11 is a schematic diagram illustrating a 3D structure of an air-conducting loudspeaker shown in FIG. 3;

[0057] FIG. 12 is an exploded schematic diagram of the air-conducting loudspeaker shown in FIG. 11;

[0058] FIG. 13 is a schematic diagram illustrating a vibration transmission sheet shown in FIG. 7 from a front view;

[0059] FIG. 14 is a schematic diagram illustrating a portion of a sectional structure of the bone-conducting loudspeaker shown in FIG. 8;

[0060] FIG. 15 is a schematic diagram illustrating an exemplary structure of an annular fixing portion of the vibration transmission sheet shown in FIG. 13;

[0061] FIG. 16 is a schematic diagram illustrating an exemplary structure of a linkage assembly of the vibration transmission sheet shown in FIG. 13 connected to a central fixing portion;

[0062] FIG. 17 is another schematic diagram illustrating a disassembled structure of the core assembly shown in FIG. 2;

[0063] FIG. 18 is a schematic diagram illustrating an exemplary structure of a bone-conducting loudspeaker shown in FIG. 17;

[0064] FIG. 19 is a schematic diagram illustrating a portion of a disassembled structure of the core assembly shown in FIG. 17;

[0065] FIG. 20 is a schematic diagram illustrating another sectional structure of the core assembly shown in FIG. 2 along B-B;

[0066] FIG. 21 is a schematic diagram illustrating an enlarged structure of an F-region shown in FIG. 20;

[0067] FIG. 22 is another schematic diagram illustrating a portion of a disassembled structure of the core assembly shown in FIG. 18;

[0068] FIG. 23 is a schematic diagram illustrating a sectional structure of the core assembly shown in FIG. 2; and

[0069] FIG. 24 is a schematic diagram illustrating a portion of a sectional structure of the core assembly shown in FIG. 23.

DETAILED DESCRIPTION

[0070] In order to help those skilled in the art better understand the technical solution provided by the present application, a further detailed description of an earphone provided by this the present application will be given below in conjunction with the accompanying drawings and specific embodiments. It should be understood that the described embodiments are merely part of the embodiments of the present application, not all embodiments. All other embodiments that can be derived by those skilled in the art without making any inventive efforts based on the embodiments in the present application fall within the scope of protection of the present application.

[0071] The terms “first”, “second”, or the like in the present application are used to distinguish between different objects and are not intended to describe a specific order. In addition, the terms “including” and “comprising”, as well as any variations thereof, are intended to cover non-exclusive inclusion. For example, a process, method, system, product, or apparatus that includes a series of steps or units is not limited to the steps or units listed but may optionally also include steps or units not listed, or may optionally include other steps or units inherent to these processes, methods, products, or apparatuses.

[0072] The following embodiments of the present disclosure describe an exemplary structure of an earphone 100.

[0073] As shown in FIG. 1, the earphone 100 may include a core assembly 1, an ear-hook assembly 2, and a rear-hook assembly 3. There may be two core assemblies 1. The two core assemblies 1 are configured to transmit vibration and/or sound to the user's left ear and right ear, respectively. The two core assemblies 1 may be the same or different. For example, one of the core assemblies 1 may be provided with a microphone, and another core assembly 1 may not be provided with a microphone. For example, one of the core assemblies 1 may be provided with a key and a corresponding circuit board, and another core assembly 1 may not be provided with a key and a corresponding circuit board. The two core assemblies 1 may be identical on a core module (e.g. a loudspeaker module). The core assembly 1 described in the following text can be considered as being detailed with one of the two core assemblies 1 as an example. There may be two ear-hook assemblies 2, and the two ear-hook assemblies 2 may be located on the user's left and right ears, respectively, so that the core assemblies 1 can be fitted to the user's face. For example, one of the ear-hook assemblies 2 may be provided with a battery, and another ear-hook assembly 2 may be provided with a control circuit, or the like. One end of the ear-hook assembly 2 is connected to the core assembly 1, and another end of the ear-hook assembly 2 is connected to the rear-hook assembly 3. The rear-hook assembly 3 connects the two ear-hook assemblies 2, and the rear-hook assembly 3 is used to wrap around the back of a user's neck or the back of a user's head and provide a clamping force that allows the two core assemblies 1 to be

clamped to two sides of a user's face and the ear-hook assemblies 2 to be more securely attached to the user's ears. It should be understood that the earphone 100 may also not include the rear-hook assembly 3, and the core assembly 1 may be worn on the user's ears through the ear-hook assembly 2.

[0074] The following content primarily provides an exemplary description of the core assembly 1 and other structures of the earphone 100.

[0075] As shown in FIG. 2, FIG. 3, FIG. 4, and FIG. 5, the core assembly 1 may include a shell assembly 10, a bone-conducting loudspeaker 11, and an air-conducting loudspeaker 12.

[0076] Optionally, the shell assembly 10 may be provided with an accommodation cavity 1001 and an accommodation cavity 1002 that are isolated from each other. The shell assembly 10 may also be referred to as a core shell assembly 10. The bone-conducting loudspeaker 11 is accommodated in the accommodation cavity 1001, and the accommodation cavity 1001 may be referred to as a first accommodation cavity. The air-conducting loudspeaker 12 is accommodated in the accommodation cavity 1002, and the accommodation cavity 1002 may be referred to as a second accommodation cavity.

[0077] The air-conducting loudspeaker 12 conducts the sound into the ear canal of the user through air vibration, and the bone-conducting loudspeaker 11 conducts the sound into the user through bone-conducting vibration. Optionally, the sealing performance of the accommodation cavity 1001 is greater than the sealing performance of the accommodation cavity 1002. The sealing performance can be considered as airtightness. Optionally, the accommodation cavity 1001 may be provided as a completely airtight accommodation cavity, and the accommodation cavity 1002 is provided as an accommodation cavity with a relatively high degree of sealing performance while guaranteeing sound generation of the air-conducting loudspeaker 12. In the above-described manner, as the accommodation cavity 1002 in which the air-conducting loudspeaker 12 is placed needs to be connected to the outside world to facilitate the conduction of the sound waves through the air, the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 are separately and independently arranged. There is no need to set the bone-conducting loudspeaker 11 in the accommodation cavity 1002. Instead, the bone-conducting loudspeaker 11 is separately arranged in the accommodation cavity 1001, and since the accommodation cavity 1001 is independent, it is possible to set the sealing performance of the accommodation cavity 1001 at a higher level, which effectively enhances the sealing effect of the bone-conducting loudspeaker 11, thereby preventing the bone-conducting loudspeaker 11 from being damaged by erosion of the external environmental factors, and at the same time guaranteeing the sound quality effect of the air-conducting loudspeaker 12. Additionally, in the earphone 100, when the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 are operating at the same time, the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 are arranged in the accommodation cavity 1001 and the accommodation cavity 1002, respectively, which can effectively reduce mutual interference between the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12, thereby effectively enhancing the sound quality of the earphone 100.

[0078] Optionally, referring to FIG. 2 and FIG. 3, the shell assembly 10 may include a shell 101, a shell 102, and a shell 103, and the shell 101 and the shell 102 fit with each other to form the accommodation cavity 1001. The shell 101 and/or the shell 102 further form a portion of the accommodation cavity 1002, and the shell 103 may form another portion of the accommodation cavity 1002, and in turn, the shell 103 and the shell 101 and/or the shell 102 fit with each other to form the accommodation cavity 1002.

[0079] The shell assembly 10 may be composed of the shell 101, the shell 102, and the shell 103. The shell 101 and the shell 102 fit with each other to form the accommodation cavity 1001, and the shell 103 and the shell 101 fit with each other to form the accommodation cavity 1002. The shell assembly 10 is formed by the shell 101, the shell 102 and the shell 103 fitting with each other in the structure described above, which can make the core assembly 1 compact and at the same time facilitate the assembly of the core assembly 1, thereby enhancing the assembly efficiency of the core assembly 1. In some embodiments, the shell 101 may be referred to as a first shell; the shell 102 may be referred to as a second shell; and the shell 103 may be referred to as a third shell.

[0080] As another example, a portion of the accommodation cavity 1002 may also be provided in the shell 102. The shell 103 and the shell 102 form the accommodation cavity 1002 by fitting with each other, or the shell 101 and the shell 102 fit with each other to form a portion of the accommodation cavity 1002, and the shell 103, the shell 101, and the shell 102 fit with each other to form the accommodation cavity 1002.

[0081] Through the above structure, it is possible to place the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 in the accommodation cavity 1001 and the accommodation cavity 1002 which are independent of each other, respectively, reducing the mutual influence between the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12. This setup maximizes the acoustic output of both the air-conducting loudspeaker 12 and the bone-conducting loudspeaker 11, while enabling a compact structure of the core assembly 1, which in turn contributes to the miniaturization of the core assembly 1 and facilitates its assembly, thereby improving the assembly efficiency of the core assembly 1.

[0082] Optionally, a vibration direction of the bone-conducting loudspeaker 11 is arranged to intersect with a vibration direction of the air-conducting loudspeaker 12, and the shell 101 and the shell 102 fit along the vibration direction of the bone-conducting loudspeaker 11. The shell 103 fits with the shell 101 and/or the shell 102 along the vibration direction of the air-conducting loudspeaker 12.

[0083] Specifically, the vibration direction of the bone-conducting loudspeaker 11 is arranged to intersect with the vibration direction of the air-conducting loudspeaker 12. The vibration direction of the bone-conducting loudspeaker 11 is referred to as a bone-conducting vibration direction X1, and the vibration direction of the air-conducting loudspeaker 12 is referred to herein as an air-conducting vibration direction X2, wherein the bone-conducting vibration direction X1 and the air-conducting vibration direction X2 are arranged to intersect rather than being parallel to each other. For example, the bone-conducting vibration direction X1 and the air-conducting vibration direction X2 are arranged perpendicularly or approximately perpendicular to each

other (e.g., $90^\circ \pm 10^\circ$). When the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** are operating at the same time, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** are vibrating along the bone-conducting vibration direction **X1** and the air-conducting vibration direction **X2**, respectively. Since the two vibration directions are arranged to intersect with each other, this can effectively mitigate the impact of the bone-conducting loudspeaker **11**'s vibration on the sound quality of the air-conducting loudspeaker **12**. Further, the shell **101** and the shell **102** may be assembled in conjunction with each other along the bone-conducting vibration direction **X1**, and the shell **103** and the shell **101** may be assembled in conjunction with each other along the air-conducting vibration direction **X2**. For example, the accommodation cavity **1002** may be formed by only the shell **101** and the shell **103** fitting together, with the shell **103** having a fitting relationship only with the shell **101** along the air-conducting vibration direction **X2**. As another example, instead of fitting with the shell **101** to form the accommodation cavity **1002**, the shell **103** may fit with other shells along the air-conducting vibration direction **X2** to form the accommodation cavity **1002**, or the shell **103** may fit with the shell **101** and other shells to form the accommodation cavity **1002**. In this way, it is favorable for the assembly of the core assembly **1** to enhance the efficiency of the assembly of the core assembly **1**.

[0084] Referring to FIG. 6 and FIG. 7, optionally, the bone-conducting loudspeaker **11** includes a voice coil assembly **111** and a magnet assembly **112**. The voice coil assembly **111** is provided around the magnet assembly **112** and is fixedly connected to the shell **101** and/or the shell **102**. The surface of the shell **101** and/or the shell **102** that is disposed opposite to the bone-conducting loudspeaker **11** along the vibration direction of the bone-conducting loudspeaker **11** is configured to contact the user's skin and transmit bone-conducting vibration.

[0085] Optionally, referring to FIG. 6, FIG. 7, and FIG. 8, the voice coil assembly **111** includes two sets of voice coils **1110** spaced apart along a central axis direction **Z** of the bone-conducting loudspeaker **11** and a magnetic conduction shield **1111** surrounding an outer periphery of the two sets of voice coils **1110**. The magnet assembly **112** includes a magnet **1121** and two magnetic conduction plates **1122**, and the two magnetic conduction plates **1122** are respectively disposed on opposite end surfaces of the magnet **1121** along the central axis direction **Z**. Projections of the two magnetic conduction plates **1122** along a radial direction of the bone-conducting loudspeaker **11** overlap with the two sets of voice coils **1110**, respectively, and current directions of the two sets of voice coils **1110** are opposite to each other. The central axis direction **Z** of the bone-conducting loudspeaker **11** coincides with the vibration direction of the bone-conducting loudspeaker **11** (i.e., the bone-conducting vibration direction **X1**).

[0086] Optionally, referring to FIG. 5, FIG. 7 and FIG. 8, an outer peripheral surface of the magnetic conduction shield **1111** may be in contact with the shell assembly **10** and/or fixedly connected to the shell assembly **10** by an adhesive material, whereby the bone-conducting loudspeaker **11** and the shell assembly **10** can be rigidly connected/rigidly contacted, such that the bone-conducting loudspeaker **11** can be securely accommodated within the accommodation cavity **1001**. Further, the outer peripheral

surface of the magnetic conduction shield **1111** may be completely in contact with the shell assembly **10** or a portion of the outer peripheral surface may be in contact with the shell assembly **10**. Further, a portion of the outer peripheral surface of the magnetic conduction shield **1111** may be partially connected to the shell assembly **10** and another portion may be fixedly connected to the shell assembly **10** by an adhesive material. For the portion in contact with the shell assembly **10**, the shell assembly **10** and/or the outer peripheral surface of the magnetic conduction shield **1111** may be provided with a structure such as a supporting rib, supporting bump, or the like, to realize the partial contact between the two.

[0087] For example, the outer peripheral surface of the magnetic conduction shield **1111** is in contact with the shell assembly **10** and/or is fixedly connected to the shell assembly **10** by an adhesive material along at least two normal directions perpendicular to each other (a normal direction of the outer peripheral surface of the magnetic conduction shield **1111**). Optionally, the outer peripheral surface of the magnetic conduction shield **1111** is in contact with the shell assembly **10** along a first direction and/or is fixedly connected to the shell assembly **10** by an adhesive material. For example, the outer peripheral surface of the magnetic conduction shield **1111** is in contact with the shell assembly **10** along a second direction and/or is fixedly connected to the shell assembly **10** by an adhesive material. For example, the first direction includes a first positive direction and a first negative direction that are opposite to each other. For example, the outer peripheral surface of the magnetic conduction shield **1111** is in contact with the shell assembly **10** and/or is fixedly connected to the shell assembly **10** by an adhesive material along the first positive direction and/or along the first negative direction. The second direction includes a second positive direction and a second negative direction that are opposite to each other. For example, the outer peripheral surface of the magnetic conduction shield **1111** is in contact with the shell assembly **10** and/or is fixedly connected to the shell assembly **10** by an adhesive material along the second positive direction and/or the second negative direction. The first direction is perpendicular to the second direction. The perpendicular mentioned here may allow for some deviation, such as angles ranging from 80° to 100° , all of which can be considered perpendicular.

[0088] As shown in FIG. 7, the outer peripheral surface of the magnetic conduction shield **1111** may include, for example, two opposite first outer side surfaces and two opposite second outer side surfaces. Each of the second outer side surfaces is located between the two first outer side surfaces, and the two second outer side surfaces and the two first outer side surfaces enclose an annular structure. For example, each of the first outer side surfaces is in contact with and/or fixedly connected to the shell assembly **10** by an adhesive material, and each of the second outer side surfaces is in contact with and/or fixedly connected to the shell assembly **10** by an adhesive material. For example, the two first outer side surfaces may be arranged opposite along the first direction, and the two second outer side surfaces may be arranged opposite along the second direction. For example, the first direction may be a direction **X2** illustrated in FIG. 5 and the second direction may be a direction **LD** illustrated in FIG. 5.

[0089] Optionally, each of the second outer side surfaces and the two first outer side surfaces may be connected

directly at an angle (corner) to each other, e.g. vertically. The transition between each second outer side surface and the two first outer side surfaces may also be curved, such as being connected by a curved connection surface. In this way, there is at least one curved connection surface between the two second outer side surfaces and the two first outer side surfaces. For example, there are four curved connection surfaces. Further, at least one curved connection surface is in contact with the shell assembly **10** or is fixedly connected to the shell assembly **10** by an adhesive material, or four curved connection surfaces are in contact with the shell assembly **10** or are fixedly connected to the shell assembly **10** by an adhesive material.

[0090] Set up in this way, a rigid contact/rigid connection can be realized between the bone-conducting loudspeaker **11** and the shell assembly **10**, effectively improving the overall strength of the shell assembly **10** and making the vibration transmission between the two more efficient, thereby reducing noises and sound leakage.

[0091] Referring to FIG. 9a, optionally, the earphone **100** includes a first working mode and a second working mode. In the first working mode, the bone-conducting loudspeaker **11** plays sound signals in a first frequency band P1, and the air-conducting loudspeaker **12** plays sound signals in a second frequency band P2. In the second working mode, the bone-conducting loudspeaker **11** plays sound signals in the first frequency band P1 and sound signals in the second frequency band P2, and the first frequency band P1 and the second frequency band P2 are at least partially offset in a frequency domain. For example, if the first frequency band P1 and the second frequency band P2 do not overlap in the frequency domain, a frequency range of the first frequency band P1 is higher than a frequency range of the second frequency band P2. For example, if the first frequency band P1 and the second frequency band P2 partially overlap in the frequency domain, a frequency range of an offset portion of the first frequency band P1 is higher than a frequency range of an offset portion of the second frequency band P2.

[0092] In some embodiments, the earphone **100** includes a frequency division point n (as shown in FIG. 9a), the first frequency band P1 includes a frequency band that is greater than the frequency division point n, and the second frequency band P2 includes a frequency band that is lower than the frequency division point n. The frequency division point n is located at an intersection point of the first frequency band P1 and the second frequency band P2. When the earphone **100** is in the first working mode, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** are operating together, the bone-conducting loudspeaker **11** mainly outputs sound signals in the first frequency band P1, and the air-conducting loudspeaker **12** mainly outputs sound signals in the second frequency band P2. That is, an operation frequency band of the bone-conducting loudspeaker **11** includes a frequency band greater than the frequency division point n, and an operation frequency band of the air-conducting loudspeaker **12** includes a frequency band lower than the frequency division point n.

[0093] Referring to FIG. 9b, FIG. 9b illustrates an electrical signal band P3 input into the bone-conducting loudspeaker **11** and an electrical signal band P4 input into the air-conducting loudspeaker **12**, respectively, when the earphone **100** is in the first working mode. In some embodiments, the electrical signal band P3 of the bone-conducting

loudspeaker **11** and the electrical signal band P2 of the air-conducting loudspeaker **12** have an overlapping intersection point m. The frequency division point n is located within a frequency range of 0.5 to 3 times the frequency span of the overlapping intersection point m. For example, if the overlapping intersection point m is 300 Hz, the frequency division point n may be located in a range of 150 Hz to 900 Hz, such as 500 Hz.

[0094] In some embodiments, the first frequency band P1 includes a mid-high frequency or a high frequency, and the second frequency band P2 includes a mid-low frequency or a low frequency. For example, a low-frequency range of 30 Hz to 150 Hz is considered a low-frequency band, a mid-frequency range of 150 Hz to 500 Hz is a mid-low frequency band, and 500 Hz to 5 KHz is a mid-high frequency band, and a high-frequency range of 5 KHz to 16 KHz is a high-frequency band.

[0095] When the earphone **100** is in the first working mode, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** are operating together, the bone-conducting loudspeaker **11** mainly outputs sound signals in the first frequency band P1, and the air-conducting loudspeaker **12** mainly outputs sound signals in the second frequency band P2. In the earphone **100**, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** operate together, wherein the bone-conducting loudspeaker **11** outputs the sound signals in the first frequency band P1 and the air-conducting loudspeaker **12** outputs the sound signals in the second frequency band P2. These signals are combined to form a more complete sound signal that is transmitted to the user's ear, which enhances the sound quality of the mid-low frequency band of the sound signal emitted by the earphone **100**, thereby significantly improving the overall sound quality of the sound signal emitted by the earphone **100**. Furthermore, the bone-conducting loudspeaker **11** may also operate separately, i.e., in the second working mode, the air-conducting loudspeaker **12** does not work, and the bone-conducting loudspeaker **11** may simultaneously output sound signals in the first frequency band P1 and sound signals in the second frequency band P2. For example, when the earphone **100** is working in an environment where the user is sweating heavily during exercise or where the user is in a humid and rainy environment, the earphone **100** may actively cut off the operation of the air-conducting loudspeaker **12**, and only the bone-conducting loudspeaker **11** is used for sound generation, thereby ensuring the overall sound quality of the sound signal emitted by the earphone **100** while preventing the air-conducting loudspeaker **12** from being eroded by unfavorable factors such as sweat or rain, effectively enhancing the reliability and stability of the earphone **100**.

[0096] Optionally, the earphone **100** may further include a third working mode. In the third working mode, the air-conducting loudspeaker **12** plays sound signals in the first frequency band P1 and the second frequency band P2. Specifically, the earphone **100** switches to the third working mode in response to detecting that the bone-conducting loudspeaker **11** is damaged and unavailable or that the environment in which the earphone **100** is currently located is unfavorable for the bone-conducting loudspeaker **11** to operate. In the third working mode, the air-conducting loudspeaker **12** operates alone and outputs sound signals in

the first frequency band P1 and the second frequency band P2, to ensure that the earphone 100 can output high-quality sound signals normally.

[0097] An alternative embodiment of the shell assembly 10 as illustrated in FIG. 10a and FIG. 10b is described below, and different from the shell assembly 10 described above, the shell assembly 10 hereinafter includes a first portion 1003 and a second portion 1004, with the first portion 1003 being connected to a wearing assembly 27 (as illustrated in FIG. 1 and FIG. 2). The second portion 1004 is detachably connected to the first portion 1003 or the wearing assembly 27. The bone-conducting loudspeaker 11 is provided on the first portion 1003, and the air-conducting loudspeaker 12 is provided on the second portion 1004 and can be removed from the wearing assembly 27 in a detachable manner. In other words, the shell assembly 10 includes the detachable first portion 1003 and the detachable second portion 1004. The bone-conducting loudspeaker 11 is provided on the first portion 1003, the air-conducting loudspeaker 12 is provided on the second portion 1004, and the second portion 1004 is detachably connected to the first portion 1003 or the wearing assembly 27. With this arrangement, the air-conducting loudspeaker 12 can be separately detachable from the rest of the components on the earphone 100, and the bone-conducting loudspeaker 11 can continue to operate after the air-conducting loudspeaker 12 is separately detached. Separately detaching the air-conducting loudspeaker 12 from the other components of the earphone 100 can enhance the environmental adaptability of the earphone 100, which allows the air-conducting loudspeaker 12 to be quickly detached from the earphone 100 in extreme environments, such as when the user is sweating heavily during exercise or in humid and rainy conditions, preventing damage to the air-conducting loudspeaker 12. Meanwhile, the earphone 100 can continue to output high-quality sound signals through the bone-conducting loudspeaker 11. When only the bone-conducting loudspeaker 11 in the earphone 100 is operating alone, the earphone 100 can control the bone-conducting loudspeaker 11 to operate through the second working mode, to ensure the sound quality of the sound signals output by the earphone 100. Furthermore, the second portion 1004 is detachably connected to the first portion 1003 or the wearing assembly 27, which facilitates the replacement of the second portion 1004. On one hand, this allows for the replacement with the air-conducting loudspeaker 12 of different performance characteristics, and on the other hand, it also makes maintenance easier.

[0098] Optionally, as shown in FIG. 10b, the earphone 100 has a wire group for supplying power and/or audio signals to, for example, the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12. The wire group may be connected within the first portion 1003. The first portion 1003 may be provided with a first electrode portion 10031, and the second portion 1004 may be provided with a second electrode portion 10041. After the first portion 1003 and the second portion 1004 are detachably assembled, the first electrode portion 10031 and the second electrode portion 10041 are in contact and can be electrically connected. The first electrode portion 10031 and the second electrode portion 10041 may have various structures, for example, the first electrode portion 10031 includes at least two first electrode contacts. The second electrode portion 10041 includes at least two second electrode contacts, and the at least two first electrode contacts and the at least two second

electrode contacts may be in one-to-one correspondence. For example, one of the first electrode portion 10031 and the second electrode portion 10041 includes a pogo pin, and another includes a leaf spring or contact. The electrical connection is established through the contact between the pogo pin and the leaf spring or contact. For example, the first electrode portion 10031 and the second electrode portion 10041 each includes a leaf spring, and the electrical connection is realized by the elastic contact between the leaf springs of the first electrode portion 10031 and the second electrode portion 10041. As another example, the first electrode portion 10031 and the second electrode portion 10041 include other forms. For example, the first electrode portion 10031 includes a plurality of ring electrodes that are nested together, and the second electrode portion 10041 includes a plurality of ring electrodes that are nested together. The first electrode portion 10031 and the second electrode portion 10041 may establish an electrical connection through the corresponding contact between their ring electrodes. The first electrode portion 10031 and the second electrode portion 10041 are not limited to the examples listed above, and the first electrode portion 10031 and the second electrode portion 10041 are configured to ensure electrical connection when the first portion 1003 and the second portion 1004 are assembled and to allow for their separation after disassembly.

[0099] In some implementations, the wire group is coupled to the bone-conducting loudspeaker 11 and the first electrode portion 10031, and the second electrode portion 10041 is coupled to the air-conducting loudspeaker 12. After the second portion 1004 and the first portion 1003 are assembled, the first electrode portion 10031 and the second electrode portion 10041 are in contact with each other correspondingly and are conductive such that the wire group is also capable of powering and/or providing audio signals to the air-conducting loudspeaker 12.

[0100] In other embodiments, the first electrode portion 10031 is coupled to the bone-conducting loudspeaker 11, and the second electrode portion 10041 is coupled to the air-conducting loudspeaker 12. The wire group is coupled to at least two first electrode portions 10031, and the wire group may supply power and/or audio signals to the bone-conducting loudspeaker 11 through the first electrode portion 10031, and may also supply power and/or audio signals to the air-conducting loudspeaker 12 through the contact between the first electrode portion 10031 and the second electrode portion 10041.

[0101] As another example, the wire group may also be connected to the second portion 1004, the first portion 1003 is detachably connected to the second portion 1004, and a specific connection relationship can be referred to the above description.

[0102] The first portion 1003 is provided with a first connection portion 10032, and the second portion 1004 is provided with a second connection portion 10042. The first connection portion 10032 and the second connection portion 10042 are detachably connected by snap-fit, adhesion, magnetic suction, threaded connection, or the like. For example, the first connection portion 10032 may be provided as one of a slot or a snap, and the second connection portion 10042 may be provided as another of a slot or a snap. The first portion 1003 and the second portion 1004 are detachably connected by a snap-fit mechanism, allowing the first elec-

trode portion **10031** and the second electrode portion **10041** to contact each other and establish an electrical connection.

[0103] Referring to FIG. 3 and FIG. 5, optionally, the shell assembly **10** may also be provided with a pressure relief hole **1081** for connecting the accommodation cavity **1002** to the external environment. The pressure relief hole **1081** is provided to extend toward the side where the accommodation cavity **1001** is located. Specifically, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** may be accommodated in the accommodation cavity **1001** and the accommodation cavity **1002**, respectively, and the shell assembly **10** is provided with the pressure relief hole **1081** for relieving pressure for the air-conducting loudspeaker **12**. The pressure relief hole **1081** enables the accommodation cavity **1002** to be in communication with the external environment, and the pressure relief hole **1081** extends from the connection position with the accommodation cavity **1002** toward the side where the accommodation cavity **1001** is located. For example, a portion of a hole wall of the pressure relief hole **1081** extends from the connection position with the accommodation cavity **1002** toward the side where the accommodation cavity **1001** is located. With this setup, it is possible to effectively reduce the shell space occupied by the pressure relief hole **1081** on a periphery of the accommodation cavity **1002**, making efficient use of the space between the accommodation cavity **1001** and the accommodation cavity **1002**. This improves the spatial utilization of the shell assembly **10**, enabling a more compact and reasonable configuration while meeting the sound production requirements of the air-conducting loudspeaker **12**. Besides, such an inclined extension can increase the dimension of the pressure relief hole **1081**, which makes it possible to provide the air-conducting loudspeaker **12** with better sound quality.

[0104] Optionally, a projection of the pressure relief hole **1081** along a direction **X3** (shown in FIG. 5) perpendicular to an arrangement direction of the accommodation cavity **1001** and the accommodation cavity **1002** at least partially overlaps with the accommodation cavity **1001**. The direction **X3** perpendicular to the arrangement direction of the accommodation cavity **1001** and the accommodation cavity **1002** may be understood as a direction perpendicular to the bone-conducting vibration direction **X1** and perpendicular to the air-conducting vibration direction **X2**. The projection of the pressure relief hole **1081** along the direction **X3** at least partially overlaps with the accommodation cavity **1001**, which ensures that the dimension of the pressure relief hole **1081** satisfies the pressure relief requirements of the air-conducting loudspeaker **12**. At the same time, this arrangement effectively reduces the space occupied by the pressure relief hole **1081** and the corresponding shell portion, thereby reducing the structural dimension of the shell assembly **10** in the region where the air-conducting loudspeaker **12** is located.

[0105] Further, the pressure relief hole **1081** is provided to extend inclined toward the side on which the accommodation cavity **1001** is located, and along the direction **X3** perpendicular to the arrangement direction of the accommodation cavity **1001** and the accommodation cavity **1002**, the dimension of the pressure relief hole **1081** gradually increases in the direction from the accommodation cavity **1002** to the external environment. In other words, while the pressure relief hole **1081** extends inclined toward the accommodation cavity **1001**, the dimension of the pressure relief

hole **1081** gradually increases in the direction from the accommodation cavity **1002** to the external environment, which makes the air discharge smoother, and can reduce the velocity of the air, thereby enhancing the sound quality of the sound emitted by the earphone **100**.

[0106] Referring to FIG. 3, FIG. 4, and FIG. 5, the shell assembly **10** includes a partition wall **1012** for isolating the accommodation cavity **1001** from the accommodation cavity **1002**. At least a portion of the partition wall **1012** extends toward the side on which the accommodation cavity **1001** is located, and forms a portion of the hole wall of the pressure relief hole **1081**.

[0107] Specifically, in a cross-section perpendicular to a vibration direction (i.e., the bone-conducting vibration direction **X1**) of the bone-conducting loudspeaker **11**, the bone-conducting loudspeaker **11** has a long-side edge **1101** and a short-side edge **1102**, with the long-side edge **1101** being longer than the short-side edge **1102**. The partition wall **1012** includes a wall segment **1013** and a wall segment **1014**. The wall segment **1013** is disposed side-by-side with the long-side edge **1101**, and the wall segment **1014** is connected to the wall segment **1013** and extends from a periphery of the connection between the long-side edge **1101** and the short-side edge **1102** toward the side on which the accommodation cavity **1001** is located. The wall segment **1014** is used to form a portion of the hole wall of the pressure relief hole **1081**. In some embodiments, the wall segment **1013** may be referred to as a first wall segment and the wall segment **1014** may also be referred to as a second wall segment.

[0108] The above spatial layout and structural arrangement of the pressure relief hole **1081** may be realized by the shell assembly **10**. It should be understood that the above spatial layout and structural arrangement of the pressure relief hole **1081** may also be realized by other solutions, which will not be discussed in detail herein. In addition to the earphone **100** provided with both the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12**, the above spatial layout and structural arrangement of the pressure relief hole **1081** may also be applied to the earphone **100** provided with only the air-conducting loudspeaker **12**, which will not be described in detail herein.

[0109] Optionally, as shown in FIG. 3, a central axis **Z** of the bone-conducting loudspeaker **11** (a central axis direction of the bone-conducting loudspeaker **11** may also be denoted by **Z**) may intersect with a central axis **V** of the air-conducting loudspeaker **12**, or may not intersect with the central axis **V** of the air-conducting loudspeaker **12**. For example, the central axis **Z** and the central axis **V** are skew lines, not lying in the same plane, and are arranged in a staggered manner. In a direction perpendicular to the central axis **Z** and the central axis **V**, the central axis **Z** is disposed farther away from the pressure relief hole **1081** compared to the central axis **V**. Generally speaking, along an LD direction shown in FIG. 5, the bone-conducting loudspeaker **11** is located farther away from the pressure relief hole **1081** compared to the air-conducting loudspeaker **12**. By setting the central axis **Z** and the central axis **V** to be skew lines and by setting the central axis **Z** farther away from the pressure relief hole **1081** as compared to the central axis **V**, a larger space can be made available for the design of the pressure relief hole **1081**, which in turn allows the dimension of the pressure relief hole **1081** to be larger, thereby enhancing the pressure relief effect.

[0110] Optionally, the distance between the central axis Z and the central axis V is in a range of 0.3 mm to 1 mm, in a range of 0.5 mm to 0.8 mm, such as 0.6 mm, 0.7 mm, or in a range of 0.2 mm to 1.2 mm. By setting the above dimensions, on the one hand, it is possible to ensure the sound quality effect of the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 while keeping the core assembly 1 small, and on the other hand, it is possible to provide a larger space for setting the pressure relief hole 1081, which in turn allows for the setting of a larger pressure relief hole 1081.

[0111] As mentioned above, the shell assembly 10 includes the shell 101, the shell 102, and the shell 103. The partition wall 1012 may be disposed on the shell 101 and/or the shell 102, and the shell 101 and the shell 102 fit with each other to form the accommodation cavity 1001 and the pressure relief hole 1081. For example, the shell 101 and/or the shell 102 further form a portion of the accommodation cavity 1002. For example, the shell 103 may form another portion of the accommodation cavity 1002, with the shell 103 fitting with the shell 101 and/or the shell 102 to form the accommodation cavity 1002. The partition wall 1012 disposed on the shell 101 and/or the shell 102 may be understood as the partition wall 1012 being a part of the shell 101 and/or the shell 102. It should be understood that how the partition wall 1012 is disposed is not limited to be disposed on the shell 101 and/or the shell 102. In other embodiments, the partition wall 1012 may be a component that is independent of the shell 101 and/or the shell 102.

[0112] As shown in FIG. 3, the partition wall 1012 is disposed on the shell 101, and the shell 101 is provided with a sub-accommodation cavity 1010 and a sub-accommodation cavity 1011 located on opposite sides of the partition wall 1012. An opening direction of the sub-accommodation cavity 1010 is oriented toward a wall surface of the partition wall 1012. Specifically, the opening direction of the sub-accommodation cavity 1010 is parallel or substantially parallel to the wall surface of the partition wall 1012. The opening direction of the sub-accommodation cavity 1011 intersects with the wall surface of the partition wall 1012. The shell 102 is provided with a sub-accommodation cavity 1020, the shell 102 is capped at an open end of the sub-accommodation cavity 1010, and the sub-accommodation cavity 1020 fits with the sub-accommodation cavity 1010 to form the accommodation cavity 1001. The shell 103 is provided with a sub-accommodation cavity 1030, the shell 103 is capped at an open end of the sub-accommodation cavity 1011, and the sub-accommodation cavity 1030 fits with the sub-accommodation cavity 1011 to form the accommodation cavity 1002. In some embodiments, the sub-accommodation cavity 1010 may also be referred to as a first sub-accommodation cavity, the sub-accommodation cavity 1011 may also be referred to as a second sub-accommodation cavity, the sub-accommodation cavity 1020 may also be referred to as a third sub-accommodation cavity, and the sub-accommodation cavity 1030 may also be referred to as a fourth sub-accommodation cavity.

[0113] By providing the partition wall 1012 to divide the shell 101 into the sub-accommodation cavity 1010 whose opening direction is oriented towards the wall surface of the partition wall 1012 and the sub-accommodation cavity 1012 that intersects with the wall surface of the partition wall 1012, both the sub-accommodation cavity 1010 and the sub-accommodation cavity 1012 can have larger spaces.

This improves the spatial utilization of the shell 101 and helps reduce mutual interference caused by vibrations between the bone-conducting loudspeaker 11 and the air-conducting loudspeaker 12 during operation, thereby enhancing sound quality.

[0114] As shown in FIG. 4, the air-conducting loudspeaker 12 includes a vibration diaphragm 123. The vibration diaphragm 123 divides the accommodation cavity 1002 into a cavity portion 1006 disposed on the side of the vibration diaphragm 123 away from the partition wall 1012 and a cavity portion 1007 disposed between the vibration diaphragm 123 and the partition wall 1012. The pressure relief hole 1081 is in communication with the cavity portion 1007. As shown in FIG. 3 and FIG. 5, the shell assembly 10 is further provided with a sound outlet hole 1080 in communication with the cavity portion 1006. The cavity portion 1007 is in communication with the external environment via the pressure relief hole 1081. When the vibration diaphragm 123 vibrates along the air-conducting vibration direction X2, the cavity portion 1007 may be relieved of the pressure via the pressure relief hole 1081, to ensure normal and stable operation of the air-conducting loudspeaker 12. In some embodiments, the cavity portion 1006 may also be referred to as a first cavity portion, and the cavity portion 1007 may also be referred to as a second cavity portion.

[0115] Optionally, the distance between the vibration diaphragm 123 and a cavity wall of the accommodation cavity 1002 along a vibration direction of the vibration diaphragm 123 (the air-conducting vibration direction X2) is greater than 0.8 mm, or greater than 1 mm. Specifically, the distance (i.e., the distance along the air-conducting vibration direction X2) between the vibration diaphragm 123 and the partition wall 1012 (i.e., a portion of the cavity wall of the accommodation cavity 1002) may be greater than 0.8 mm, or greater than 1 mm. This ensures that the vibration diaphragm 123 has a larger vibration space, and the distance enables the airflow in the cavity portion 1007 to be distributed more uniformly, which in turn enables the air pressure to be more uniform and improves the consistency of the vibration of the vibration diaphragm 123. In some embodiments, the distance between the vibration diaphragm 123 and the partition wall 1012 refers to the minimum distance between the two. In other embodiments, the distance between the vibration diaphragm 123 and the partition wall 1012 refers to the distance between a vibration region of the vibration diaphragm 123 (e.g., a center region or a region of maximum amplitude) and the partition wall 1012.

[0116] As shown in FIG. 3 to FIG. 5, optionally, in a wearing state, the center of the pressure relief hole 1081 is not higher than the center of the sound outlet hole 1080, and in the wearing state, a line connecting the center of the pressure relief hole 1081 and the center of the sound outlet hole 1080 is pointed toward the user's ear. Optionally, in the wearing state, the line connecting the center of the pressure relief hole 1081 to the center of the sound outlet hole 1080 is pointed toward the ear canal of the user's ear. Generally speaking, the line connecting the center of the pressure relief hole 1081 and the center of the sound outlet hole 1080 may be inclined upward toward the head top direction relative to the human body and point toward the user's ear. The center of the pressure relief hole 1081 may be the centroid of the pressure relief hole 1081. The center of the sound outlet hole 1080 may be the centroid of the sound outlet hole 1080. This setup helps to avoid the user's ear (especially the ear canal)

being at the acoustic null point in the wearing state or in cases of improper wearing (such as slight tilting). As a result, it ensures that the user can hear the sound properly, reducing hearing loss and improving sound quality.

[0117] If there is a plurality of sound outlet holes 1080, the line described above may be a line connecting the center of at least one of the sound outlet holes 1080 and the center of the pressure relief hole 1081. As another example, if there is a plurality of sound outlet holes 1080, a line connecting the centers of the plurality of sound outlet holes 1080 forms a shape, and the center of the shape serves as the center of the plurality of sound outlet holes 1080, and the line described above may be a line connecting the center of the shape and the center of the pressure relief hole 1081 that is pointed toward the user's ear.

[0118] Optionally, in the wearing state, an angle between the line connecting the center of the pressure relief hole 1081 and the center of the sound outlet hole 1080 and the coronal axis of the user is $\geq 0^\circ$ and $< 90^\circ$, and an angle between the line and the vertical axis of the user is $> 0^\circ$ and $\leq 90^\circ$. This setup helps to avoid the user's ear being at the acoustic null point in the wearing state or in cases of improper wearing (such as slight tilting). As a result, it ensures the output of the earphone 100 and the acoustic effect.

[0119] As shown in FIG. 4, the air-conducting loudspeaker 12 further includes a magnetic circuit assembly 120 and a voice coil 122. The magnetic circuit assembly 120 includes a shield 121 with an open end 1211, and an annular flange 1212 that is disposed at the open end 1211 of the shield 121 and protrudes from an outer peripheral surface of the shield 121. A rim of the vibration diaphragm 123 is fixed to a side of the annular flange 1212 facing the open end 1211 and the vibration diaphragm 123 covers the open end 1211 of the shield 121. The voice coil 122 is connected to the vibration diaphragm 123. As shown in FIG. 3 and FIG. 5, the vibration diaphragm 123 is arranged opposite to the shield 121, away from the sound outlet hole 1080.

[0120] The air-conducting loudspeaker 12 requires a larger vibration diaphragm 123 to achieve good sound quality, and the larger vibration diaphragm 123 also requires a larger support structure such as the annular flange 1212. In other words, the annular flange 1212 and the vibration diaphragm 123 are both structures in the air-conducting loudspeaker 12 that have large radial dimensions compared to the other structures, which greatly affects the overall structural dimension of the shell assembly 10. By setting the annular flange 1212 and the vibration diaphragm 123 on the side away from the sound outlet hole 1080, the vibration diaphragm 123 and the annular flange 1212 can be located closer to the interior of the shell assembly 10. With such an arrangement, in the air-conducting loudspeaker 12, structures with a larger radial dimension are arranged away from the sound outlet hole 1080, and structures with a smaller dimension can be located closer to the sound outlet hole 1080. Such an opposite configuration inside the air-conducting loudspeaker 12, compared to the traditional loudspeaker design where the vibration diaphragm must be oriented toward the sound outlet hole 1080, effectively reduces the dimension of the portion of the shell assembly 10 that is near the sound outlet hole 1080, which allows the dimension of the shell assembly 10 to decrease from the central region to the region around the sound outlet hole 1080 (i.e., the radial dimension of the outer peripheral surface of the shell assembly 10 can gradually decrease). As a result, the structure

becomes more compact, significantly improving the spatial utilization of the shell assembly 10 and reducing the overall volume of the shell assembly 10. Such an opposite configuration can optimize the sound path and thus improve the sound quality. In short, by reversing the air-conducting loudspeaker 12 along the air-conducting vibration direction X2, the structural dimension of the shell assembly 10 can be effectively reduced.

[0121] Optionally, as shown in FIG. 3 and FIG. 4, the cavity wall of the accommodation cavity 1002 is provided with an annular table surface 1005 (the annular table surface 1005 is also shown in FIG. 5, but not labeled). As shown in FIG. 4 and FIG. 5, the annular flange 1212 is pressed against the annular table surface 1005. The vibration diaphragm 123 divides the accommodation cavity 1002 into the cavity portion 1006 and the cavity portion 1007, with the cavity portion 1006 being disposed on the side of the vibration diaphragm 123 facing the sound outlet hole 1080 and the cavity portion 1007 being disposed on the side of the vibration diaphragm 123 away from the sound outlet hole 1080. The magnetic circuit assembly 120 is located within the cavity portion 1006. As shown in FIG. 5, the shell assembly 10 is provided with the pressure relief hole 1081 to enable communication between the cavity portion 1007 and the external environment. Specifically, the annular table surface 1005 may be provided on shell 101.

[0122] As shown in FIG. 4, the vibration diaphragm 123 may be disposed on the side of the annular flange 1212 toward the annular table surface 1005. The earphone 100 further includes a sealing adhesive 1041 that seals a gap between the shell assembly 10 and the side of the annular flange 1212 that is opposite the annular table surface 1005. The annular flange 1212 fits with the shell assembly 10 to form an adhesive accommodation groove 1042 for accommodating the sealing adhesive 1041 on the side of the annular flange 1212 that is opposite the annular table surface 1005. Based on reversing the air-conducting loudspeaker 12, since the vibration diaphragm 123 is closer to an inner side of the shell assembly 10 compared to the adhesive accommodation groove 1042, while the adhesive accommodation groove 1042 is closer to an outer side of the shell assembly 10, on the one hand, it is easy to dispense the adhesive, and on the other hand, when dispensing the adhesive on the adhesive accommodation groove 1042, it can prevent the adhesive from overflowing onto the vibration diaphragm 123, reducing the effect of the adhesive on the vibration diaphragm 123, thereby ensuring the sound quality effect.

[0123] Further as shown in FIG. 11, a sound-conducting hole 1213 is disposed on the annular flange 1212. The sound-conducting hole 1213 is provided to connect a space surrounded by the vibration diaphragm 123 and the magnetic circuit assembly 120 to the cavity portion 1006 around the periphery of the magnetic circuit assembly 120. A portion of sound waves generated by the vibration of the vibration diaphragm 123 along the air-conducting vibration direction X2 is conducted via the sound-conducting hole 1213 to the cavity portion 1006 around the periphery of the magnetic circuit assembly 120.

[0124] Optionally, as shown in FIG. 4, the shell 103 is used to accommodate one end of the shield 121 away from the vibration diaphragm 123, and the sound outlet hole 1080 is provided on the shell 103. The shield 121 includes a circumferential side surface 121a and an end surface 121b, and the end surface 121b is connected to the circumferential

side surface **121a**. The end surface **121b** is oriented toward the sound outlet hole **1080**. A first gap **1031** (a position of the dashed line pointed to by the label **1031** in FIG. 4) exists between a connection region between the end surface **121b** and the circumferential side surface **121a** located on a first side of the shield **121** and the shell **103**. A second gap **1032** (a position of the dashed line pointed to by the label **1032** in FIG. 4) exists between a connection region between the circumferential side surface **121a** located on a second side of the shield **121** and the end surface **121b** and the shell **103**. The first gap **1031** is smaller than the second gap **1032**. The size of the first gap **1031** and the size of the second gap **1032** may be distances between the corresponding connection regions to the shell **103**, respectively. For example, the distance is the length of the dashed line illustrated in FIG. 4 (the dashed line may be perpendicular to an inner surface of the shell **103** at a corresponding location). The first side and the second side refer to a first side and a second side of the circumferential side surface **121a** that are opposite to each other.

[0125] Referring to FIG. 4, FIG. 5, FIG. 11, and FIG. 12, in a cross-section perpendicular to the vibration direction of the vibration diaphragm **123**, the shield **121** has a length direction LD and a width direction HD. The dimension of the shield **121** along the length direction LD is larger than the dimension of the shield **121** along the width direction HD, and the first side and the second side are opposite along the width direction HD. The first side may be located on the side of the magnetic circuit assembly **120** close to the user's face, as shown in FIG. 4. The first gap **1031** is smaller than the second gap **1032**, and the first gap **1031** is located on the first side, which in turn makes the smaller first gap **1031** closer to the user's face, and can ensure that the air-conducting loudspeaker **12** outputs sound signals normally, while also effectively reducing the dimension of the shell **103**.

[0126] Further, the first side is located on the side of the magnetic circuit assembly **120** close to the user's face, and at least a portion of an outer surface of the shell **103** is provided inclined toward the shell **101** and the shell **102** along a direction from the second side to the first side. For example, the portion of the outer surface of the shell provided with the sound outlet hole **1080** is provided inclined toward the shell **101** and the shell **102** along a direction from the second side to the first side, i.e., the sound outlet hole **1080** is disposed on the at least a portion of the shell **103**. With this setting manner, it is possible to make the sound outlet hole **1080** toward the user's ear, so that the sound signal output from the air-conducting loudspeaker **12** can be better transmitted into the user's ear. A portion of the shell **103** that is opposite the circumferential side surface **121a** is connected to a portion of the shell provided with the sound outlet hole. Optionally, after reversing the air-conducting loudspeaker **12**, an outer surface of the portion of the shell **103** that is opposite the circumferential side surface **121a** may be provided in an inclined manner, and specifically, the outer surface may gradually converge and tilt along the air-conducting vibration direction X2, from the first shell **101** to the direction away from the first shell **101** and into the accommodation cavity **1002**, to make the shell assembly **10** compact and miniaturized, thereby reducing the volume of the shell assembly **10**. Moreover, the shell assembly **10** may have a fitting surface **10a** in contact with the user's face and an inclined surface **10b** (the inclined surface

may be a curved inclined surface that gradually tilts away from the face) connected to the fitting surface **10a** and tilted away from the fitting surface **10a**. The inclined surface **10b** is used to avoid the user's tragus, thereby reducing the compression on the tragus, facilitating wearing, improving the wearing comfort, and having better sound quality effect. The fitting surface **10a** is mainly composed of the shell **102**, and the inclined surface **10b** may be composed of the shell **102** and the shell **103** together.

[0127] Further, a portion of the end surface **121b** adjacent to the circumferential side surface **121a** disposed on the first side may be connected to the shell **103**, or a connection region between the end surface **121b** and the circumferential side surface **121a** disposed on the first side may be in contact with the shell **103**, further reducing the dimension of the shell assembly **10**. The contact here can be direct contact, where further adhesive bonding can be applied for fixation, or it can be indirect contact through adhesive fixation. A portion of the cavity portion **1006** that is on the periphery of the shield **121** may be further divided into a first space portion and a second space portion by the above contact position. Specifically, the first space portion may be enclosed between the circumferential side surface **121a** disposed on the first side and the shell **103**, or the first space portion may be enclosed by a combination of the end surface **121b**, the circumferential side surface **121a** disposed on the first side, and the shell **103**, while the second space portion may be enclosed between at least a portion of the end surface **121b**, a portion of the circumferential side surface **121a** disposed on the second side, and the shell **103**. The first gap **1031** is located in the first space portion, the second gap is located in the second space portion, and the second space portion is in communication with the sound outlet hole **1080**. If the end surface **121b** encloses both the first space portion and the second space portion, the area of the portion of the end surface **121b** enclosing the first space portion is smaller than the area of the portion of the end surface **121b** enclosing the second space portion, i.e., the contact position is closer to the first side and to the portion of the shell assembly **10** opposite the first side. Corresponding positions of the shield **121** and the shell **103** may be contacted to form the contact position. In other embodiments, the corresponding positions of the shield **121** and the shell **103** may be set to have a gap whose distance is greater than 0.1 mm, and the gap is so set that the shield **121** does not collide with the shell **103** when vibrates due to the vibration of the vibration diaphragm **123**, thereby reducing the possibility of generating noise. In conclusion, corresponding to the contact position, a position corresponding to the gap is referred to as a minimum gap position.

[0128] In the state in which the contact position is set, at least a portion of sound waves can enter the second space portion from the first space portion along the circumferential side surface **121a**, and then pass out from the second space portion via the sound outlet hole **1080**.

[0129] Optionally, referring to FIG. 4, on the first side, the average distance between the annular flange **1212** and the shell **103** along the vibration direction X2 of the vibration diaphragm **123** is greater than or equal to 0.5 mm. Optionally, on the second side, the second gap **1032** is greater than or equal to 0.5 mm. On the first side, the average distance between the annular flange **1212** and the shell **103** along the vibration direction X2 of the vibration diaphragm **123** is the average value of the sum of the maximum distance L5 and

the minimum distance L4 between the annular flange 1212 and the shell 103 along the vibration direction of the vibration diaphragm 123 (i.e., the air-conducting vibration direction X2).

[0130] By setting the above dimensions, on the one hand, it is ensured that there is a sufficient size of a sound cavity to achieve a better sound quality before leading the sound waves out of the sound outlet hole 1080, and on the other hand, it is possible to make the shell assembly 10 smaller and more compact. In addition, by reversing the accommodation cavity 1002 and the air-conducting loudspeaker 12 as shown in FIG. 4, the sound waves enter the second space portion from the first space portion along the circumferential side surface 121a (i.e., the sound waves can enter the second gap 1032 from the first gap 1031), and along the air-conducting vibration direction X2, the second space portion is gradually inclined to the first space portion to better point to the sound outlet hole 1080, so that the transmission direction of the sound waves is more directed to the user's ear, further enhancing the transmission efficiency of the sound waves, and enhancing the sound quality.

[0131] Optionally, as shown in FIG. 4, the shield 121 includes a magnetic conduction shield 121c. The magnetic circuit assembly 120 further includes the magnet 124 disposed within the magnetic conduction shield 121c, and the voice coil 122 extends into a magnetic gap formed by the magnetic conduction shield 121c and the magnet 124.

[0132] Referring to FIG. 6, FIG. 7, and FIG. 8, the bone-conducting loudspeaker 11 includes the voice coil assembly 111, the magnet assembly 112, and a vibration transmission sheet 113. The first one of the voice coil assembly 111 and the magnet assembly 112 surrounds the second one of the voice coil assembly 111 and the magnet assembly 112. For example, the voice coil assembly 111 surrounds the magnet assembly 112. As shown in FIG. 8 and FIG. 13, the vibration transmission sheet 113 includes a central fixing portion 1131, an annular fixing portion 1132 around the periphery of the central fixing portion 1131, and a linkage assembly 1133 connected between the central fixing portion 1131 and the annular fixing portion 1132. The annular fixing portion 1132 is connected to the first one of the voice coil assembly 111 and the magnet assembly 112, and the central fixing portion 1132 is connected to the second one of the voice coil assembly 111 and the magnet assembly 112. For example, the annular fixing portion 1132 is connected to the voice coil assembly 111 and the central fixing portion 1131 is connected to the magnet assembly 112. The linkage assembly 1133 is used to elastically constrain relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z of the bone-conducting loudspeaker 11. It should be understood that the linkage assembly 1133 can maintain the stable position of the voice coil assembly 111 and the magnet assembly 112 along a radial direction perpendicular to the central axis direction Z, i.e., the linkage assembly 1133 can constrain the voice coil assembly 111 and the magnet assembly 112 along a radial direction. The relatively stable position of the voice coil assembly 111 and the magnet assembly 112 along the radial direction can render the relative movement between the two along the central axis direction Z more stable, which in turn makes the sound quality better. For example, the annular fixing portion 1132 serves as a stop member and rigidly constrains a range of relative movement between the voice coil assembly 111 and

the magnet assembly 112 along the central axis direction Z (illustrated in FIG. 6). For example, the annular fixing portion 1132 is fixedly connected to the voice coil assembly 111. When used as a stop member, the magnet assembly 112 may be rigidly stopped when the vibration magnitude of the magnet assembly 112 relative to the voice coil assembly 111 is excessive, thereby limiting the movement stroke of the magnet assembly 112 and imposing a rigid constraint on the range of relative movement between the voice coil assembly 111 and the magnet assembly 112. It should be understood that the stop member may be realized in other structural forms or ways.

[0133] The central axis direction Z of the bone-conducting loudspeaker 11 may coincide with the vibration direction of the bone-conducting loudspeaker 11 (i.e., the bone-conducting vibration direction X1). The linkage assembly 1133 is an elastic member or can generate an elastic deformation that elastically constrains the voice coil assembly 111 and the magnet assembly 112. The elastic constraint may be understood as the voice coil assembly 111 and the magnet assembly 112 moving relative to each other within the range of relative movement permitted by the elastic deformation of the linkage assembly 1133. On the one hand, the linkage assembly 1133 can limit the range of relative movement between the voice coil assembly 111 and the magnet assembly 112 along the bone-conducting vibration direction X1, and on the other hand, the linkage assembly 1133 can reset the voice coil assembly 111 and the magnet assembly 112 through elastic recovery after the voice coil assembly 111 and the magnet assembly 112 have achieved relative movement. Rigidly constraining the range of relative movement between the voice coil assembly 111 and the magnet assembly 112 by the stop member along the central axis direction Z may be understood as the stop member constraining the maximum range of relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z, preventing the relative movement between the voice coil assembly 111 and the magnet assembly 112 from exceeding the maximum range of relative movement and causing the vibration transmission sheet 113 to transition from elastic deformation to plastic deformation, resulting in a decrease in the elasticity of the vibration transmission sheet 113 and causing damage to the voice coil assembly 111 and the magnet assembly 112. Rigidity and elasticity herein are relative, i.e., the constraining strength of the stop member is greater than the constraining strength of the linkage assembly 1133.

[0134] As shown in FIG. 8, the annular fixing portion 1132 serves as a stop member, and a projection of the annular fixing portion 1132 along the central axis direction Z partially overlaps with an outer end surface 1103 of the second one of the voice coil assembly 111 and the magnet assembly 112. Specifically, the annular fixing portion 1132 is connected to the voice coil assembly 111, and the projection of the annular fixing portion 1132 along the central axis direction Z partially overlaps an outer end surface 1103 of the magnet assembly 112. By setting the projection of the annular fixing portion 1132 to partially overlap the outer end surface 1103 of the second one of the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z, when the relative movement amplitude of the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z is too large, the outer end surface 1103 of the voice coil assembly 111 and the magnet assembly

bly 112 may abut against with the annular fixing portion 1132. Since the annular fixing portion 1132 is fixed relative to the first one of the voice coil assembly 111 and the magnet assembly 112, the annular fixing portion 1132 rigidly abuts against the second one of the voice coil assembly 111 and the magnet assembly 112, thus rigidly constraining the range of relative movement between the voice coil assembly 111 and the magnet assembly 112. For example, if the earphone 100 is subjected to the drop or collision, the relative movement amplitude of the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z may be too excessive, resulting in the vibration transmission sheet 113 transition from elastic deformation to plastic deformation, which reduces its elasticity. By applying a rigid constraint through the annular fixing portion 1132, the vibration transmission sheet 113 can be prevented from being overstretched, thereby mitigating performance degradation and reducing the impact on sound quality. In addition, providing the annular fixing portion 1132 as the stop member can eliminate the need to add an additional stopper structure, leading to a simple and stable structure with reduced costs.

[0135] Optionally, as shown in FIG. 6, FIG. 7, and FIG. 8, the outer end surface 1103 includes a center region 1104 and a peripheral region 1106 surrounding the center region 1104. The center region 1104 is provided with a protrusion portion 1105, and the protrusion portion 1105 protrudes with respect to the peripheral region 1106. The central fixing portion 1131 is fixedly connected to the protrusion portion 1105, and the projection of the annular fixing portion 1132 along the central axis direction Z partially overlaps with the peripheral region 1106.

[0136] Optionally, referring to FIG. 14, along the radial direction of the bone-conducting loudspeaker 11, a width L6 of an overlapping region between the annular fixing portion 1132 and the outer end surface 1103 is greater than or equal to 0.03 mm, thereby ensuring the constraining strength of the stop member.

[0137] Optionally, in a natural state, a spacing distance L7 between an upper surface of the protrusion portion 1105 and a lower surface of the annular fixing portion 1132 of the vibration transmission sheet 113 along the central axis direction Z is less than or equal to 1 mm. Setting the spacing distance L7 within the above range enables the linkage assembly 1133 to be reasonably pre-deformed when the central fixing portion 1131 and the protrusion portion 1105 are fixedly connected, which in turn improves the vibration efficiency of the vibration transmission sheet 113. If the spacing distance L7 is set too large, the pre-deformation amount of the linkage assembly 1133 will be excessive, making it prone to exceeding the yield limit and undergoing plastic deformation, which reduces its elastic deformation performance. Conversely, with the vibration transmission sheet 113 elastically suspended, the absence of pre-deformation would cause the relative position of the magnet assembly 112 and the voice coil assembly 111 along the central axis direction Z to deviate from the central position under the influence of gravity. This deviation would affect the relative movement between the magnet assembly 112 and the voice coil assembly 111, thereby affecting sound quality performance. In some embodiments, since the lower surface of the central fixing portion 1131 and the upper surface of the protrusion portion 1105 need to be welded in affinity, the spacing distance L7 between the upper surface of the protrusion portion 1105 and the lower surface of the

annular fixing portion 1132 along the central axis direction Z may also be considered as the spacing distance between the lower surface of the central fixing portion 1131 and the lower surface of the annular fixing portion 1132.

[0138] It should be understood that whether the protrusion portion 1105 is fixed to the central fixing portion 1131 by some other structures or means, the spacing distance between the lower surface of the central fixing portion 1131 and the lower surface of the annular fixing portion 1132 along the central axis direction Z indicates the pre-deformation amount of the vibration transmission sheet 113 along the central axis direction Z.

[0139] Optionally, in a natural state, a ratio of the spacing distance between the lower surface of the central fixing portion 1131 and the lower surface of the annular fixing portion 1132 along the central axis direction Z to an extension component of the length of at least one linkage 1134 along a first direction W1 is in a range of 0 to 0.1, and the range of values includes endpoint values. Optionally, a ratio of the spacing distance between the lower surface of the central fixing portion 1131 and the lower surface of the annular fixing portion 1132 along the central axis direction Z to the extension component of the length of the at least one linkage 1134 along a second direction W2 is in a range of 0 to 0.18, and the range of values includes endpoint values. With such an arrangement, it is possible to make the vibration transmission sheet 113 be capable of pre-deformation reasonably. On the one hand, suspended components (the voice coil assembly 111 or the magnet assembly 112) can be ensured sufficiently supported without sagging to ensure the vibration performance, and on the other hand, an excellent space for elastic deformation can be guaranteed. If the ratio is too large, it is likely to result in an excessive pre-deformation amount of the vibration transmission sheet 113 that exceeds the yield limit, resulting in a decrease in elastic performance. If the ratio is too small, it is likely to lead to insufficient support of the vibration transmission sheet 113 to the suspended components, which leads to degradation of vibration performance and loss of sound quality.

[0140] Optionally, the upper surface of the protrusion portion 1105 occupies less than or equal to 30% of the area of the outer end surface 1103, and the protrusion portion 1105 occupying a smaller area can realize a more stable fixed connection with the linkage assembly 1133. In this way, it does not occupy the linkage assembly 1133 excessively, but makes the linkage assembly 1133 have a sufficiently large area to generate elastic deformation, thereby providing sufficient deformation space to make the vibration transmission sheet 113 more elastic.

[0141] Optionally, as shown in FIG. 7 and FIG. 8, the magnet assembly 112 includes the magnet 1121 and two magnetic conduction plates 1122. The two magnetic conduction plates 1122 are disposed on two opposite end surfaces of the magnet 1121 along the central axis direction Z. The protrusion portion 1105 is disposed on the magnetic conduction plates 1122 such that the center region 1104 and the peripheral region 1106 are formed on an outer surface of each of the magnetic conduction plates 1122. Two projections of the two magnetic conduction plates 1122 along a radial projection of the bone-conducting loudspeaker 11 overlap with the voice coil assembly 111, respectively. The peripheral region 1106 is provided affixed to the magnet 1121, the protrusion portion 1105 is disposed spaced apart

from the magnet 1121, and the peripheral region 1106 occupies more than or equal to 60% of the area on the outer end surface 1103. In this way, setting the protrusion portion 1105 can make the pre-deformation amount of the vibration transmission sheet 113 more reasonable, and can reduce the pre-deformation amount, ensuring the elastic deformation performance of the vibration transmission sheet 113. In addition, there is also no need to set up additional connection members to connect the vibration transmission sheet 113 and the magnetic conduction plate 1122, which can reduce the overall mass of the entire bone-conducting loudspeaker 11, and thus can improve the vibration effect of the bone-conducting loudspeaker 11. Optionally, when the bone-conducting loudspeaker 11 is in a normal operation state, the stop member is provided so as not to constrain the relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z. However, when the bone-conducting loudspeaker 11 is subjected to dropping, collision, or the like, the relative movement between the voice coil assembly 111 and the magnet assembly 112 may exceed the elastic constraint range permitted by the elastic deformation of the vibration transmission sheet 113, then at this time, the stop member may rigidly constrain the relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z, reducing the probability that the relative movement between the voice coil assembly 111 and the magnet assembly 112 exceeds the permitted elastic constraint range, thereby reducing the probability of loss of elasticity of the vibration transmission sheet 113 due to overstretching.

[0142] Optionally, as shown in FIG. 7 and FIG. 8, there are two vibration transmission sheets 113, and the two vibration transmission sheets 113 are provided on two opposite sides of the bone-conducting loudspeaker 11 along the central axis direction Z. The linkage assembly 1133 of each vibration transmission sheet 113 includes two linkages 1134. In other embodiments, the linkage assembly 1133 also includes at least two linkages 1134.

[0143] Optionally, as shown in FIG. 7 and FIG. 8, the voice coil assembly 111 surrounds the magnet assembly 112. The voice coil assembly 111 includes two sets of voice coils 1110 spaced apart along the central axis direction Z and the magnetic conduction shield 1111 around the periphery of the two sets of voice coils 1110. The magnet assembly 112 includes the magnet 1121 and two magnetic conduction plates 1122. The two magnetic conduction plates 1122 are provided on two opposite end surfaces of the magnet 1121 along the central axis direction Z. Projections of the two magnetic conduction plates 1122 along the radial direction of the bone-conducting loudspeaker 11 overlap with the two sets of voice coils 1110, respectively, and current directions of the two sets of voice coils 1110 are opposite to each other.

[0144] Optionally, the voice coil assembly 111 surrounds the magnet assembly 112, and the voice coil assembly 111 is fixed to the shell assembly 10.

[0145] Optionally, the annular fixing portion 1132 is fixedly connected to the magnetic conduction shield 1111. Optionally, the annular fixing portion 1132 and the magnetic conduction shield 1111 are welded and fixed through offset welding. Specifically, an outer edge of the annular fixing portion 1132 is offset from an outer edge of a corresponding end portion of the magnetic conduction shield 1111. For example, the outer edge of the annular fixing portion 1132

overlaps the end portion of the magnetic conduction shield 1111 and is located between the outer peripheral surface and the inner peripheral surface of the magnetic conduction shield 1111, thereby forming an offset. Based on this offset, the outer edge of the annular fixing portion 1132 and the end portion of the magnetic conduction shield 1111 are welded and fixed. By welding and fixing the annular fixing portion 1132 and the magnetic conduction shield 1111 through offset welding, the welding position is more obviously presented, which facilitates the welding process and observation of the weld seam, effectively improving welding efficiency and yield rate, thereby ensuring welding strength. For the form of welding the weld seam, it may be spot welding or continuous welding. Optionally, the weld bearing force between the annular fixing portion 1132 and the magnetic conduction shield 1111 is in a range of 400N to 1200N.

[0146] It may be appreciated that in some other embodiments of the present disclosure, the welding method may also be other methods such as lap welding (butt welding). The lap welding specifically involves stacking the annular fixing portion 1132 with the corresponding end portion of the magnetic conduction shield 1111, and welding the stacked position together using methods such as laser welding.

[0147] Optionally, the vibration transmission sheet 113 is provided with an observation hole 1130 penetrating through two sides of the vibration transmission sheet 113, as shown in FIG. 13 and FIG. 15. The observation hole 1130 penetrating through two sides of the vibration transmission sheet 113 refers to that the observation hole 1130 penetrates through two sides of the vibration transmission sheet 113 along a thickness direction of the vibration transmission sheet 113, and the thickness direction of the vibration transmission sheet 113 may be the central axis direction Z. Specifically, the observation hole 1130 may be formed in the annular fixing portion 1132. Optionally, there are a plurality of observation holes 1130 arranged at intervals along the circumferential direction of the vibration transmission sheet 113. By providing the observation hole 1130 on the vibration transmission sheet 113, a structure of a magnetic circuit, such as a structure of the magnet assembly 112, inside the bone-conducting loudspeaker 11 can be observed through the observation hole 1130. Besides, the observation hole 1130 can be used to understand whether the position of the magnet assembly 112 is deviated, and even the working condition of the magnet assembly 112 can be observed, which is convenient for fault troubleshooting or structural optimization, or the like.

[0148] Optionally, the vibration transmission sheet 113 includes the central fixing portion 1131 formed by sheet processing, the annular fixing portion 1132 around the periphery of the central fixing portion 1131, and the linkage assembly 1133 connected between the central fixing portion 1131 and the annular fixing portion 1132. An inner ring edge 1132a of the annular fixing portion 1132 has the first direction W1 and the second direction W2 that are perpendicular to each other. That is, the annular fixing portion 1132 has the first direction W1 and the second direction W2. The linkage assembly 1133 is composed of two linkages 1134, each of the linkages 1134 has an outer edge 1135 toward the inner ring edge 1132a, and an inner edge 1136 away from the inner ring edge 1132a. An extension component of the outer edge 1135 along the first direction W1 is greater than or equal to one-half of a dimension of the inner ring edge 1132a

along the first direction W1, and the width of the linkage 1134 is greater than or equal to one-tenth of a dimension of the inner ring edge 1132a along the second direction W2.

[0149] Specifically, the above description is illustrated in a case where the central fixing portion 1131, the annular fixing portion 1132, and the linkage assembly 1133 are set up coplanar. In the case where there are distortion points such as protrusions or grooves at the inner ring edge 1132a, these distortion points should be eliminated, and a smooth linear transition should be formed on the outside of the distortion points to create the inner ring edge 1132a. For example, the second direction W2 refers to a direction in which the shortest line segment intersecting the inner ring edge 1132a over the geometric center of the inner ring edge 1132a is located. The first direction W1 refers to a direction that is perpendicular to the second direction W2 in the plane in which the central fixing portion 1131, the annular fixing portion 1132, and the linkage assembly 1133 are located. The vibration transmission sheet 113 is provided as a two-linkage structure, and an extension component of the outer edge 1135 along the first direction W1 is greater than or equal to one-half of the dimension of the inner ring edge 1132a along the first direction W1. In this way, the extension length of the linkage 1134 can be increased, which makes the linkage 1134 has a larger deformation space. At the same time, since the width of the linkage 1134 is greater than or equal to one-tenth of the dimension of the inner ring edge 1132a along the second direction W2, it can improve the resistance as well as the service life of the linkage 1134.

[0150] Optionally, the extension component of the outer edge 1135 along the first direction W1 is further greater than or equal to two-thirds of the dimension of the inner ring edge 1132a along the first direction W1. Therefore, the extension component of the outer edge 1135 along the first direction W1 is further greater than or equal to two-thirds of the dimension of the inner ring edge 1132a along the first direction W1, which may further increase the extension length of the linkage 1134 and allows the linkage 1134 to have a greater deformation space.

[0151] Optionally, the extension component of the outer edge 1135 along the second direction W2 is greater than or equal to one-half of the dimension of the inner ring edge 1132a along the second direction W2. Therefore, the extension component of the outer edge 1135 along the second direction W2 is greater than or equal to one-half of the dimension of the inner ring edge 1132a along the second direction W2, which may further increase the extension length of the linkage 1134 and allows the linkage 1134 to have a larger deformation space.

[0152] Optionally, the extension component of the outer edge 1135 along the second direction W2 is greater than or equal to two-thirds of the dimension of the inner ring edge 1132a along the second direction W2. Therefore, the extension component of the outer edge 1135 along the second direction W2 is greater than or equal to two-thirds of the dimension of the inner ring edge 1132a along the second direction W2, which can further increase the extension length of the linkage 1134 and allows the linkage 1134 to have a larger deformation space.

[0153] Optionally, the width of the linkage 1134 is greater than or equal to one-eighth of the dimension of the inner ring edge 1132a along the second direction W2. Therefore, the width of the linkage 1134 is greater than or equal to one-eighth of the dimension of the inner ring edge 1132a

along the second direction W2, which can further improve the resistance as well as the service life of the linkage 1134.

[0154] Optionally, the two linkages 1134 are rotationally symmetrical at 180 degrees centered on the central fixing portion 1131. With this setup, the vibration transmission stability and balance of the vibration transmission sheet 113 can be effectively improved.

[0155] Optionally, the dimension of the inner ring edge 1132a along the first direction W1 is larger than the dimension of the inner ring edge 1132a along the second direction W2. In other words, the annular fixing portion 1132 may be provided in a general runway shape. A ratio of the dimension of the inner ring edge 1132a along the first direction W1 to the dimension of the inner ring edge 1132a along the second direction W2 is greater than or equal to 1.5.

[0156] Optionally, along the first direction W1, two linkages 1134 are connected to two opposite sides of the central fixing portion 1131. With this setup, it is possible to increase the length of the linkages 1134 effectively. Specifically, as shown in FIG. 15, the inner ring edge 1132a includes two edge segments 11321 spaced apart along the second direction W2. Each of the linkages 1134 is provided to start at the central fixing portion 1131, to extend through a gap between the central fixing portion 1131 and one of the two inner ring edges 1132a, and to be connected to the other of the two edge segments 11321.

[0157] Optionally, one end of the linkage 1134 is smoothly connected to the central fixing portion 1131. Another end of the linkage 1134 is smoothly connected to the annular fixing portion 1132. Further, as shown in FIG. 16, the linkage 1134 includes a first sub-linkage portion 1137 connected to the central fixing portion 1131, a second sub-linkage portion 1138 connected to the annular fixing portion 1132 and a third sub-linkage portion 1139 connected between the first sub-linkage portion 1137 and the second sub-linkage portion 1138. At least a portion of the first sub-linkage portion 1137 is progressively wider in a direction toward the central fixing portion 1131 and smoothly connected to the central fixing portion 1131, at least a portion of the second sub-linkage portion 1138 is progressively wider in a direction toward the annular fixing portion 1132 and smoothly connected to the annular fixing portion 1132, and the third sub-linkage portion 1139 remains uniform in width.

[0158] Optionally, the inner edge 1136 of the first sub-linkage portion 1137 and the inner edge 1136 of the second sub-linkage portion 1138 are provided in an arcuate shape. The inner edge 1136 of the third sub-linkage portion 1139 is provided in a straight line and is in smooth transition with the inner edge 1136 of the first sub-linkage portion 1137 and the inner edge 1136 of the second sub-linkage portion 1138, and/or the outer edge 1135 of the first sub-linkage portion 1137 and the outer edge 1135 of the second sub-linkage portion 1138 are provided in an arcuate shape, and the outer edge 1135 of the third sub-linkage portion 1139 is provided in a straight line and is in a smooth transition with the outer edge 1135 of the first sub-linkage portion 1137 and the outer edge 1135 of the second sub-linkage portion 1138. Such a setting can effectively reduce the internal stress at the connection point between the linkage 1134 and the central fixing portion 1131 and the connection point between the linkage 1134 and the annular fixing portion 1132.

[0159] Optionally, a portion of the second sub-linkage portion 1138 that is connected to the third sub-linkage portion 1139 is aligned with the width of the third sub-

linkage portion 1139 and/or a portion of the first sub-linkage portion 1137 that is connected to the third sub-linkage portion 1139 is aligned with the width of the third sub-linkage portion 1139.

[0160] Optionally, an extension component of the outer edge 1135 of the third sub-linkage portion 1139 along the first direction W1 is greater than or equal to one-half of the dimension of the inner ring edge 1132a along the first direction W1, and the width of the third sub-linkage portion 1139 is greater than 0.5 mm.

[0161] Optionally, the inner edge 1136 of the first sub-linkage portion 1137 is provided as a concave arc, and an edge of the central fixing portion 1131 that is connected to the inner edge 1136 of the first sub-linkage portion 1137 is provided as a convex arc, forming a smooth transition with the inner edge 1136 of the first sub-linkage portion 1137. Set up in this way, on the one hand, it is possible to make a reasonable transition between the central fixing portion 1131 and the first sub-linkage portion 1137, and on the other hand, the convex arc setting can ensure the dimension of the central fixing portion 1131, and thus ensure the fixing effect of the central fixing portion 1131 and the magnet assembly 112.

[0162] Optionally, the bone-conducting loudspeaker 11 includes two vibration transmission sheets 113. The two vibration transmission sheets 113 are provided on two opposite sides of the bone-conducting loudspeaker 11 along the central axis direction Z, respectively. The extension direction of the linkage assembly 1133 of the first one of the two vibration transmission sheets 113, starting from the connected central fixing portion 1131, is opposite to the extension direction of the linkage assembly 1133 of the second one of the two vibration transmission sheets 113, which also starts from the connected central fixing portion 1131. With this setup, the moment of inertia around the bone-conducting vibration direction X1 can be offset when the bone-conducting loudspeaker 11 vibrates along the bone-conducting vibration direction X1.

[0163] Optionally, the magnet assembly 112 is less than or equal to 80% of the mass of the bone-conducting loudspeaker 11, and the mass of the load pushed by the bone-conducting loudspeaker 11 is less than or equal to 9 times the mass of the bone-conducting loudspeaker 11. The load pushed by the bone-conducting loudspeaker 11 includes the sum of the masses of the shell assembly 10 and the parts within the shell assembly 10 other than the magnet assembly 112.

[0164] Optionally, the mass of the bone-conducting loudspeaker 11 is less than or equal to 8 grams.

[0165] Optionally, the vibration transmission sheet 113 is set such that the earphone 100 has a resonance peak, with the peak of the resonance peak being less than or equal to 500 Hz. Referring to FIG. 3 and FIG. 17, the core assembly 1 includes the shell assembly 10, the bone-conducting loudspeaker 11, and a fixing adhesive 1062. The shell assembly 10 is used to accommodate the bone-conducting loudspeaker 11. The fixing adhesive 1062 is used to fix the bone-conducting loudspeaker 11 to the shell assembly 10. Specifically, the shell assembly 10 is provided with the accommodation cavity 1001. The bone-conducting loudspeaker 11 may be accommodated within the accommodation cavity 1001, and the bone-conducting loudspeaker 11 is provided to vibrate along the central axis direction Z of the bone-conducting loudspeaker 11. The portion of the shell assembly

10 corresponding to the bone-conducting loudspeaker 11 is fitted to the user's face, and the bone-conducting loudspeaker 11 can drive the shell assembly 10 to vibrate when vibrating, converting an electrical signal into a vibration signal and transmitting the vibration signal to the auditory nerve through the human tissues. The central axis direction Z of the bone-conducting loudspeaker 11 is shown in direction Z in FIG. 17.

[0166] As shown in FIG. 3 and FIG. 17, the earphone 100 further includes the wearing assembly 27. The shell 101 and the shell 102 fit with each other to form the accommodation cavity 1001, the shell 101 is connected to the wearing assembly 27, and the fixing adhesive 1062 connects the shell 101 to the bone-conducting loudspeaker 11. Fitting the shell 101 and the shell 102 to form the accommodation cavity 1001 can facilitate the earphone 100 to be installed with the wearing assembly 27 and the bone-conducting loudspeaker 11 during production, reduce the difficulty of the production of the earphone 100, and also facilitate the subsequent disassembly and maintenance of the earphone 100. The core assembly 1 includes the shell assembly 10 and a microphone assembly 13. The microphone assembly 13 is accommodated within the accommodation cavity 1001. As shown in FIG. 17, the shell assembly 10 is provided with a sound inlet hole 1082, and the microphone assembly 13 is configured to capture an external sound input via the sound inlet hole 1082.

[0167] For example, the wearing assembly 27 includes elements such as an ear-hook, a headband, a clip, or the like, to facilitate wearing and fixing the earphone 100 in a position via the wearing assembly 27. Optionally, the wearing assembly 27 is disposed on the shell 101 along a radial direction of the bone-conducting loudspeaker 11 such that the wearing assembly 27 does not interfere with the vibration effect brought about by the bone-conducting loudspeaker 11 along the central axis direction Z.

[0168] As shown in FIG. 7, the voice coil assembly 111 surrounds the magnet assembly 112, and the vibration transmission sheet 113 is elastically connected to the voice coil assembly 111 and the magnet assembly 112 to elastically constrain the relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z. The fixing adhesive 1062 connects the shell assembly 10 and the voice coil assembly 111.

[0169] Specifically, the voice coil assembly 111 may interact with the magnet assembly 112 while current is passing through the voice coil assembly 111, thereby generating vibration. Therefore, by controlling the current passing through the voice coil assembly 111, the voice coil assembly 111 may interact with the magnet assembly 112 to generate corresponding vibration signals, and the vibration signals may be further transmitted to the shell assembly 10 and then to human tissues.

[0170] The vibration transmission sheet 113 is elastically connected to the voice coil assembly 111 and the magnet assembly 112 and can elastically constrain the voice coil assembly 111 and the magnet assembly 112, so that when the voice coil assembly 111 generates vibration, the voice coil assembly 111 and the magnet assembly 112 are less likely to move relative to each other along the central axis direction Z, thereby ensuring the stability of the earphone 100.

[0171] Optionally, the fixing adhesive 1062 connects the shell 101 and the voice coil assembly 111. Specifically, the inner peripheral surface of the shell 101 and an outer

peripheral surface of the voice coil assembly 111 may be opposite to each other, and the fixing adhesive 1062 may be filled in a gap between the inner peripheral surface of the shell 101 and the outer peripheral surface of the voice coil assembly 111 to connect and fix the shell 101 and the voice coil assembly 111.

[0172] Optionally, the shell 101 and the shell 102 may fit along the central axis direction Z, and the shell 102 further presses and fixes the voice coil assembly 111 to the shell 101. In this way, by using the shell 102 to further press and fix the voice coil assembly 111 to the shell 101, it is possible to further stabilize the position of the voice coil assembly 111 along the central axis direction Z. The shell 101 and shell 102 fitting to further press and fix the voice coil assembly 111 reduces the need for additional fixing means, and also makes it easier for the voice coil assembly 111 to drive the shell assembly 10 to vibrate when it vibrates along the central axis direction Z, thereby improving the structural tightness of the earphone 100.

[0173] As previously described, there may be two sets of voice coils 1110, with the magnetic conduction shield 1111 around the periphery of the two sets of voice coils 1110. The two sets of voice coils 1110 are available for the passage of current for vibration by interacting with the magnet assembly 112 as the current passes. Besides, providing the two sets of voice coils 1110 spaced apart along the central axis direction Z enables a stronger vibration of the voice coil assembly 111.

[0174] The magnetic conduction shield 1111 is used to constrain the magnetic field direction of the magnet assembly 112, and is also used to contact the shell assembly 10. When the two sets of voice coils 1110 vibrate, the magnetic conduction shield 1111 can be driven to vibrate to transmit the vibration signals to the shell assembly 10 through the magnetic conduction shield 1111.

[0175] As previously described, there may be two magnetic conduction plates 1122, and the two magnetic conduction plates 1122 may be disposed on two opposite end surfaces of the magnet 1121 along the central axis direction Z, respectively. The two magnetic conduction plates 1122 are configured to constrain the magnetic field direction of the magnet 1121 on two opposite end surfaces of the magnet 1121 along the central axis direction Z, thereby enhancing the effect of the interaction between the magnet 1121 and the voice coils 1110. Optionally, projections of the two magnetic conduction plates 1122 along a radial direction E overlap with the two voice coils 1110, ensuring that the magnetic conduction plates 1122 can act on the two voice coils 1110. The current directions of the two sets of voice coils 1110 are opposite to ensure that the two voice coils 1110 are subjected to the same direction of force under the interaction of the same magnet 1121.

[0176] A locating portion 1061 may be provided on the shell assembly 10 and/or the bone-conducting loudspeaker 11 as illustrated in FIG. 17 to FIG. 19. The locating portion 1061 is configured to maintain a predetermined gap between an inner peripheral surface of the shell assembly 10 and an outer peripheral surface of the bone-conducting loudspeaker 11 along a radial direction of the bone-conducting loudspeaker 11. The radial direction of the bone-conducting loudspeaker 11 is shown in the direction E in FIG. 17 to FIG. 19. Directions perpendicular to the central axis direction Z are all referred to as the radial direction E of the bone-conducting loudspeaker 11.

[0177] The position of the locating portion 1061 may be disposed on the shell assembly 10, on the bone-conducting loudspeaker 11, or on both the shell assembly 10 and the bone-conducting loudspeaker 11, which is not limited by the present disclosure herein. For example, the locating portion 1061 may be disposed on one of the shell assembly 10 or the bone-conducting loudspeaker 11, and is positioned against another one of the shell assembly 10 or the bone-conducting loudspeaker 11 along a radial direction of the bone-conducting loudspeaker 11, thus forming a gap between the inner peripheral surface of the shell assembly 10 and the outer peripheral surface of the bone-conducting loudspeaker 11.

[0178] The fixing adhesive 1062 may fill in the gap and connect the shell assembly 10 and the bone-conducting loudspeaker 11. This allows the fixing adhesive 1062 to fix the shell assembly 10 and the bone-conducting loudspeaker 11 along a circumferential direction of the bone-conducting loudspeaker 11, so that the bone-conducting loudspeaker 11 is less prone to deflection when vibrating along the central axis direction Z.

[0179] Maintaining the predetermined gap between the bone-conducting loudspeaker 11 and the shell assembly 10 along the circumferential direction of the bone-conducting loudspeaker 11 facilitates the addition of the fixing adhesive 1062, simplifies the production process of the earphone 100, and facilitates the flow of the fixing adhesive 1062 along the circumferential direction of the bone-conducting loudspeaker 11, which enables it to be uniformly filled in the gap between the bone-conducting loudspeaker 11 and the shell assembly 10. In this way, the fixing effect of the fixing adhesive 1062 can be improved, thus stabilizing the position of the bone-conducting loudspeaker 11 in the accommodation cavity 1001, thereby making the structure of the earphone 100 more compact and stable.

[0180] Optionally, there may be a plurality of locating portions 1061, and the plurality of locating portions 1061 are spaced apart along the circumferential direction of the bone-conducting loudspeaker 11. The plurality of locating portions 1061 are spaced apart along the circumferential direction of the bone-conducting loudspeaker 11, and the plurality of locating portions 1061 locate predetermined gaps along the circumferential direction of the bone-conducting loudspeaker 11 to facilitate application of the fixing adhesive 1062 along the circumferential direction of the bone-conducting loudspeaker 11, thereby enhancing the fixation effect of the fixing adhesive 1062. Optionally, the plurality of locating portions 1061 may be symmetrically arranged on two opposite sides of the bone-conducting loudspeaker 11 along the circumferential direction.

[0181] As shown in FIG. 18, the vibration transmission sheet 113 may elastically constrain the relative movement between the voice coil assembly 111 and the magnet assembly 112 along the central axis direction Z. Optionally, the central fixing portion 1131 is fixedly connected to the magnetic conduction plate 1122, the annular fixing portion 1132 is fixedly connected to the magnetic conduction shield 1111, and the linkage assembly 1133 connects the annular fixing portion 1132 and the central fixing portion 1131.

[0182] Specifically, the locating portion 1061 is disposed at an outer ring edge of the annular fixing portion 1132 of the vibration transmission sheet 113 and extends along the radial direction of the bone-conducting loudspeaker 11 toward the inner peripheral surface of the shell assembly 10 for abutting against the inner peripheral surface of the shell assembly 10,

such that a predetermined gap can be maintained between the inner peripheral surface of the shell assembly 10 and the outer peripheral surface of the bone-conducting loudspeaker 11 along the radial direction of the bone-conducting loudspeaker 11, enabling the fixing adhesive 1062 to be filled within the gap. By disposing the locating portion 1061 on the vibration transmission sheet 113 can facilitate the machining out of the locating portion 1061 through a simple machining process. Besides, the vibration of the vibration transmission sheet 113 is mainly along the central axis direction Z and relies on elastic deformation generated by the linkage assembly 1133, on the one hand, disposing the locating portion 1061 at the outer ring edge of the annular fixing portion 1132 may not affect the elastic deformation of the linkage assembly 1133, ensuring the vibration and sound quality effect of the bone-conducting loudspeaker 11, and on the other hand, it can also enhance the strength of the annular fixing portion 1132, and the smaller displacement and the relatively fixed position of the annular fixing portion 1132 can better support the inner peripheral surface of the shell assembly 10 and can more stably maintain the predetermined gap.

[0183] As previously described, there may be two vibration transmission sheets 113, and the two vibration transmission sheets 113 are provided on two opposite sides of the bone-conducting loudspeaker 11 along the central axis direction Z. The locating portions 1061 may be disposed at the outer ring edge of the annular fixing portion 1132 of the two vibration transmission sheets 113, respectively, which can make the force of the bone-conducting loudspeaker 11 more uniform. Along the central axis direction Z of the bone-conducting loudspeaker 11, the gap between the inner peripheral surface of the shell assembly 10 and the outer peripheral surface of the bone-conducting loudspeaker 11 may be of a consistent width, which can enable the bone-conducting loudspeaker 11 to be set in a more accurate position and also enhance the filling effect of the fixing adhesive 1062.

[0184] In some embodiments, the locating portion 1061 is disposed at the outer ring edge of the annular fixing portion 1132 of the vibration transmission sheet 113, which also improves the assembly and positioning of the vibration transmission sheet 113 when it is fixed to the magnet assembly 112 and the magnetic conduction shield 1111 with a better locating accuracy, which in turn ensures that a magnetic gap between the magnetic conduction shield 1111 fixed with the voice coil 1110 and the magnet assembly 112 more uniform and balanced, thereby improving the sound quality. For example, the bone-conducting loudspeaker 11 can be assembled using an upper jig and a lower jig as follows: the magnet assembly 112 and the magnetic conduction shield 1111 fixed with the voice coil 1110 are located using the lower jig, while the vibration transmission sheet 113 is located using the upper jig. In this way, the vibration transmission sheet 113 can be located using the locating portion 1061, and the upper jig can be designed with holes to facilitate the welding of the vibration transmission sheet 113 to the magnet assembly 112 and the magnetic conduction shield 1111, which also ensures the alignment of corresponding positions on the vibration transmission sheet 113 (e.g., the central fixing portion 1131) with the corresponding holes on the upper jig. Then, the upper jig and the lower jig are aligned to ensure that the corresponding positions on the vibration transmission sheet 113 (e.g., the central fixing

portion 1131) align with the corresponding positions on the magnet assembly 112 (e.g., the protrusion portion 1105). Similarly, the corresponding positions on the vibration transmission sheet 113 (e.g., the annular fixing portion 1132) align with the corresponding positions on the magnetic conduction shield 1111. This alignment allows for subsequent operations through the holes in the upper jig, such as pre-deforming the vibration transmission sheet 113 and fixing the vibration transmission sheet 113, the magnetic conduction shield 1111, and the magnet assembly 112 together.

[0185] As shown in FIG. 19 to FIG. 21, the locating portion 1061 may also be integrally molded to the inner peripheral surface of the shell assembly 10.

[0186] In some embodiments, a plurality of locating portions 1061 may be spaced apart on an inner peripheral surface of the shell assembly 10. The plurality of locating portions 1061 may be spaced apart on the shell assembly 10, extend towards the bone-conducting loudspeaker 11, and abut against the outer peripheral surface of the bone-conducting loudspeaker 11 to support a gap between the inner peripheral surface of the shell assembly 10 and the outer peripheral surface of the bone-conducting loudspeaker 11. Optionally, the plurality of locating portions 1061 may be integrally molded to the inner peripheral surface of the shell 101. By providing the locating portions 1061 on the inner peripheral surface of the shell assembly 10, on the one hand, it can locate the bone-conducting loudspeaker 11 when assembling the bone-conducting loudspeaker 11, which facilitates its assembly, on the other hand, since the locating portion 1061 is disposed on the inner peripheral surface of the shell assembly 10, its position is stable and is not affected by the vibration of the bone-conducting loudspeaker 11, and thus the predetermined gap can be stably maintained. Moreover, by disposing the locating portion 1061 on the shell assembly 10, the bone-conducting loudspeaker 11 can be minimized and facilitate the vibration of the bone-conducting loudspeaker 11, and by setting the locating portion 1061 on the shell assembly 10 abutting against the bone-conducting loudspeaker 11, the vibration transmission of the bone-conducting loudspeaker 11 to the shell assembly 10 can be further ensured, thereby ensuring the sound quality effect of the bone-conducting loudspeaker 11. Moreover, locating on the shell assembly 10 is relatively simple to process, which can streamline the manufacturing process of the locating portion 1061 and, in turn, reduce the overall manufacturing complexity of the earphone 100.

[0187] In other embodiments, as illustrated in FIG. 22, the locating portion 1061 includes a first locating portion 1063 integrally molded to the outer ring edge of the annular fixing portion 1132 and a second locating portion 1064 integrally molded to the inner peripheral surface of the annular fixing portion 1132. The first locating portion 1063 is disposed on the annular fixing portion 1132 and extends along the radial direction E of the bone-conducting loudspeaker 11 to abut against the inner peripheral surface of the shell assembly 10. The second locating portion 1064 is disposed on the inner peripheral surface of the shell assembly 10 and extends along the radial direction E of the bone-conducting loudspeaker 11 to abut against the outer peripheral surface of the bone-conducting loudspeaker 11. It should be understood that the first locating portion 1063 may not be limited to being disposed on the annular fixing portion 1132, but may also be disposed on the outer peripheral surface of the

bone-conducting loudspeaker **11** such as the outer peripheral surface of the voice coil assembly **111**.

[0188] Both the bone-conducting loudspeaker **11** and the shell assembly **10** are provided with the locating portions **1061** to realize a bidirectional abut along the radial direction **E**, so that even if the locating portion **1061** on one of the sides is not effective in supporting the bone-conducting loudspeaker **11**, the locating portion **1061** on another side can ensure the support effect, which in turn enhances the redundancy and further stabilizes the position of the bone-conducting loudspeaker **11**. In this way, the locating portion **1061** can more securely support the gap between the inner peripheral surface of the shell assembly **10** and the outer peripheral surface of the bone-conducting loudspeaker **11**, to facilitate the addition of the fixing adhesive **1062** and to make the force of the bone-conducting loudspeaker **11** more even, thus making the structure of the earphone **100** more stable.

[0189] Optionally, the first locating portion **1063** and the second locating portion **1064** are staggered from each other along the circumferential direction of the bone-conducting loudspeaker **11**. Such an arrangement facilitates the assembly of the bone-conducting loudspeaker **11** into the shell assembly **10**, reduces friction and damage caused by the first locating portion **1063** and the second locating portion **1064** due to coming into contact with each other, and further stabilizes the position of the bone-conducting loudspeaker **11** and enhance the structural tightness of the earphone **100**.

[0190] Referring to FIG. **23**, the shell assembly **10** includes the accommodation cavity **1001**. The bone-conducting loudspeaker **11** is accommodated within the accommodation cavity **1001**. The vibration transmission sheet **113** may connect the voice coil assembly **111** and the magnet assembly **112** and elastically constrain the relative movement between the voice coil assembly **111** and the magnet assembly **112** along the central axis direction **Z** of the bone-conducting loudspeaker **11**. First one of the voice coil assembly **111** and the magnet assembly **112** is rigidly connected to the shell assembly **10**, which in turn can transmit vibrations generated by the relative movement between the voice coil assembly **111** and the magnet assembly **112** to the shell assembly **10** for realizing bone conduction.

[0191] Specifically, the first one of the voice coil assembly **111** and the magnet assembly **112** refers to the voice coil assembly **111** and the second one of the voice coil assembly **111** and the magnet assembly **112** refers to the magnet assembly **112**. In this case, the voice coil assembly **111** may be rigidly connected to the shell assembly **10**. In other embodiments, the first one of the voice coil assembly **111** and the magnet assembly **112** also refers to the magnet assembly **112**, and the second one of the voice coil assembly **111** and the magnet assembly **112** refers to the magnet assembly **112**.

[0192] As previously described, the central fixing portion **1131** may be connected to the magnet assembly **112** and the annular fixing portion **1132** may be connected to the voice coil assembly **111**. The vibration transmission sheet **113** may be inherently elastic or may be subject to elastic deformation. The vibration transmission sheet **113** may adaptively deform elastically when the voice coil assembly **111** and the magnet assembly **112** move relative to each other. While the vibration transmission sheet **113** adaptively deforms elastically, the central fixing portion **1131** connecting the magnet

assembly **112** and the annular fixing portion **1132** connecting the voice coil assembly **111** may move relative to each other along the central axis direction **Z**.

[0193] Usually, when the earphone **100** outputs sound normally, the range of relative movement between the magnet assembly **112** and the voice coil assembly **111** is within a permitted range, i.e., the relative movement between the two is realized through the elastic deformation generated by the vibration transmission sheet **113**. However, in some special scenarios, the range of relative movement between the magnet assembly **112** and the voice coil assembly **111** may exceed the permitted range. For example, when the earphone **100** falls to the ground or collide, the magnet assembly **112** continues to move due to inertia, resulting in an excessive displacement of the magnet assembly **112** relative to the voice coil assembly **111**, thereby exerting excessive force on the vibration transmission sheet **113**. Consequently, the vibration transmission sheet **113** may be overstretched, potentially exceed its range of elastic deformation, and undergo plastic deformation, which would reduce its elasticity, weaken the elastic constraint effect, and ultimately affect the performance of the earphone **100**.

[0194] To ameliorate the above-mentioned problems, the shell assembly **10** may be provided with the stop member **107**. The stop member **107** may rigidly constrain the range of relative movement between the voice coil assembly **111** and the magnet assembly **112** along the central axis direction **Z**. That is to say, the stop member **107** may apply a rigid constraint to the magnet assembly **112** when the magnet assembly **112** moves excessively compared to the voice coil assembly **111**, to limit the movement amplitude of the magnet assembly **112**, which in turn can reduce the elasticity degradation of the vibration transmission sheet **113** caused by the excessive movement amplitude.

[0195] Setting the stop member **107** to limit the range of relative movement between the voice coil assembly **111** and the magnet assembly **112** can avoid an excessively large range of relative movement between the voice coil assembly **111** and the magnet assembly **112** or reduce the possibility of an excessively large range of relative movement between the voice coil assembly **111** and the magnet assembly **112**, thereby minimizing the possibility of overstretching the vibration transmission sheet **113**.

[0196] The rigid constraint imposed by the stop member **107** and the elastic constraint imposed by the vibration transmission sheet **113** are relative to each other, i.e., the constraining strength exerted by the stop member **107** on the magnet assembly **112** when stopping the magnet assembly **112** is greater than the constraining strength exerted by the vibration transmission sheet **113** on the magnet assembly **112** and the voice coil assembly **111**.

[0197] Optionally, the stop member **107** protrudes from the shell assembly **10**. The first one of the voice coil assembly **111** and the magnet assembly **112** may be fixedly connected to the shell assembly **10**, and the stop member **107** is disposed on a surface of the shell assembly **10** towards the second one of the voice coil assembly **111** and the magnet assembly **112** along the central axis direction **Z**. The stop member **107** may be fixedly connected to the second one of the voice coil assembly **111** and the magnet assembly **112** or abut against the vibration transmission sheet **113** after the second one of the voice coil assembly **111** and the magnet assembly **112** moves a predetermined distance relative to the first one of the voice coil assembly **111**

and the magnet assembly 112, which in turn can effectively limit the range of relative movement between the voice coil assembly 111 and the magnet assembly 112.

[0198] For example, the voice coil assembly 111 is fixedly connected to the shell assembly 10, and the stop member 107 may abut against the magnet assembly 112 or the vibration transmission sheet 113 after the magnet assembly 112 moves to a predetermined distance relative to the voice coil assembly 111, thereby imposing a rigid constraint on the magnet assembly 112 or the vibration transmission sheet 113. This makes it possible to limit the range of relative movement between the voice coil assembly 111 and the magnet assembly 112.

[0199] In other embodiments, the stop member 107 may also be in the form of a plate or other arbitrary shape.

[0200] Optionally, the stop member 107 is integrally molded to the shell assembly 10, which can improve productivity and assembly efficiency.

[0201] The stop member 107 may apply a rigid constraint to the magnet assembly 112 directly or indirectly after the magnet assembly 112 moves a predetermined distance relative to the voice coil assembly 111. Optionally, the stop member 107 applies the rigid constraint to the magnet assembly 112 indirectly by applying the rigid constraint directly on the vibration transmission sheet 113.

[0202] Referring to FIG. 24, the stop member 107 may abut against the central fixing portion 1131 of the vibration transmission sheet 113 and/or the linkage assembly 1133 to apply a rigid constraint on the vibration transmission sheet 113.

[0203] Optionally, the stop member 107 includes a first stop member 1071 for abutting against the central fixing portion 1131. The first stop member 1071 may effectively limit the vibration range of the vibration transmission sheet 113 and thus limit the range of relative movement between the voice coil assembly 111 and the magnet assembly 112.

[0204] Optionally, the stop member 107 includes a second stop member 1072 for abutting against the linkage assembly 1133. The second stop member 1072 may fit with the first stop member 1071 to allow the stop member 107 and the vibration transmission sheet 113 to be abutted against each other to a greater extent, so as to more contact the vibration transmission sheet 113 and further protect the vibration transmission sheet 113.

[0205] Optionally, there may be at least one first stop member 1071. Optionally, there may be at least two second stop members 1072.

[0206] During vibration of the vibration transmission sheet 113, the moveable distance of a region near the center of the vibration transmission sheet 113 (e.g., the central fixing portion 1131) is greater than the moveable distance of a region away from the center of the vibration transmission sheet 113. If the first stop member 1071 and the second stop member 1072 are at the same distance from the bone-conducting loudspeaker 11 while the movement degrees in different regions of the vibration transmission sheet 113 are different, the first stop member 1071 and the second stop member 1072 may not simultaneously abut against the vibration transmission sheet 113, which makes the rigid constraint less effective.

[0207] Optionally, the vibration transmission sheet 113 may be configured with a pre-deformation design, i.e., in a natural state, the linkage assembly 1133 has an extension component starting at the annular fixing portion 1132 and

pointing toward the interior of the bone-conducting loudspeaker 11 along the central axis direction Z. The vibration transmission sheet 113 may be pre-deformed toward the interior of the bone-conducting loudspeaker 11 or may be pre-deformed toward the exterior of the bone-conducting loudspeaker 11. The natural state refers to a state when the bone-conducting loudspeaker 11 is assembled or a state in which the voice coil assembly 111 and the magnet assembly 112 are relatively stationary when the bone-conducting loudspeaker 11 is not operating.

[0208] Along the central axis direction Z, a spacing d2 from the first stop member 1071 to the central fixing portion 1131 is greater than a spacing d1 from the second stop member 1072 to the linkage assembly 1133. In this way, the possibility of the first stop member 1071 and the second stop member 1072 simultaneously abutting against the vibration transmission sheet 113 can be increased, which facilitates the first stop member 1071 and the second stop member 1072 to jointly exert a rigid constraint on the vibration transmission sheet 113. Alternatively, as compared to the spacing d2 from the first stop member 1071 to the central fixing portion 1131 being equal to the spacing d1 from the second stop member 1072 to the linkage assembly 1133, the spacing d2 from the first stop member 1071 to the central fixing portion 1131 being greater than the spacing d1 from the second stop member 1072 to the linkage assembly 1133 can provide sufficient vibration space for the vibration transmission sheet 113 to vibrate during normal operation, thereby enhancing the sound quality of the earphone 100.

[0209] It should be understood that in a natural state, the second stop member 1072 and the linkage assembly 1133 may be spaced apart from each other or may abut against each other. Since the linkage assembly 1133 is relatively weak in rigidity and is susceptible to stretching, the second stop member 1072 and the linkage assembly 1133 abutting against each other in the natural state can further limit the vibration degree of the vibration transmission sheet 113 and better protect the linkage assembly 1133.

[0210] Optionally, the spacing d2 from the first stop member 1071 to the central fixing portion 1131 is greater than or equal to 0.1 mm, which can make the first stop member 1071 more sensitive against the vibration transmission sheet 113 and provide timely protection. In other words, the bone-conducting loudspeaker 11 can achieve bone conduction relatively well while being less prone to damage during drops or collisions.

[0211] Considering that the spacing d1 from the second stop member 1072 to the linkage assembly 1133 is smaller than the spacing d2 from the first stop member 1071 to the central fixing portion 1131, if the second stop member 1072 is too close to the first stop member 1071 (the central fixing portion 1131), this may limit the moveable distance of the magnet assembly 112 (i.e., the range of relative movement between the voice coil assembly 111 and the magnet assembly 112). Optionally, along an extension direction of the linkage assembly 1133, the position of the second stop member 1072 abutting against the linkage assembly 1133 is closer to the annular fixing portion 1132 as compared to the central fixing portion 1131. In this way, the vibration degree of the central fixing portion 1131 can be retained to a certain extent, which allows the voice coil assembly 111 and the magnet assembly 112 of the bone-conducting loudspeaker 11 to have a larger range of relative movement within a reasonable limit.

[02112] As described previously, the outer end surface 1103 of the second one of the voice coil assembly 111 and the magnet assembly 112 includes the center region 1104 and the peripheral region 1106 surrounding the center region 1104 (refer to FIG. 7 and FIG. 8 for illustrations). The center region 1104 is provided with the protrusion portion 1105. The protrusion portion 1105 protrudes with respect to the peripheral region 1106. The central fixing portion 1131 is connected to the protrusion portion 1105. In this case, it is advantageous for the second one of the voice coil assembly 111 and the magnet assembly 112 and the central fixing portion 1131 to be assembled together.

[02113] Taking the outer end surface 1103 disposed on the magnet assembly 112 as an example, the outer end surface 1103 includes the center region 1104 and the peripheral region 1106. By disposing the protrusion portion 1105 on the outer end surface 1103 of the magnet assembly 112, and fixedly connecting the central fixing portion 1131 of the vibration transmission sheet 113 to the protrusion portion 1105, on the one hand, this configuration minimizes the fixed connection region between the vibration transmission sheet 113 and the magnet assembly 112, reducing the limitations on the elastic deformation of the vibration transmission sheet 113 and improving the vibration effect while ensuring the elastic constraint, on the other hand, by not disposing the protrusion portion 1105 on the vibration transmission sheet 113, the weight of the vibration transmission sheet 113 can be kept at a certain level to reduce its impact on the vibration. At the same time, this design allows for a more reliable and stable connection between the magnet assembly 112 and the vibration transmission sheet 113.

[02114] Specifically, when the central fixing portion 1131 of the vibration transmission sheet 113 is rounded, the protrusion portion 1105 may also be cylindrical. The shape of the center region 1104 can be designed to match the shape of the central fixing portion 1131, ensuring that a stable connection is formed between the center region 1104 and the central fixing portion 1131 of the vibration transmission sheet 113.

[02115] Optionally, as shown in FIG. 17, the shell assembly 10 includes the shell 101 and the shell 102 that fit with each other along the central axis direction Z. The stop member 107 is provided on an inner surface of the shell 101 and/or an inner surface of the shell 102. The outer surface of the shell 101 and/or the outer surface of the shell 102 is in contact with the user's skin. The assembly of the bone-conducting loudspeaker 11 is facilitated by the above structure.

[02116] Optionally, the first one of the voice coil assembly 111 and the magnet assembly 112 surrounds the second one of the voice coil assembly 111 and the magnet assembly 112. For example, the voice coil assembly 111 surrounds the magnet assembly 112. The voice coil assembly 111 is rigidly connected to the shell 101 and/or the shell 102. By setting the voice coil assembly 111 surrounding the magnet assembly 112, it can minimize the magnet assembly 112, which can reduce the volume of the bone-conducting loudspeaker 11, and thus reduce the volume of the earphone 100. In other embodiments, the magnet assembly 112 may also be provided outside the voice coil assembly 111.

[02117] Optionally, the shell 102 further presses and fixes the voice coil assembly 111 to the shell 101, so that the voice coil assembly 111 can be stably sandwiched between the

shell 101 and the shell 102 with a better fixing effect, and thus enhancing the effect of vibration conduction to the shell assembly 10 and easily assembling the bone-conducting loudspeaker 11 in the shell assembly 10.

[02118] The foregoing is only a part of the embodiments of the present disclosure, and is not intended to limit the scope of protection of the present disclosure, and any equivalent device or equivalent process transformations utilizing the contents of the present disclosure and the accompanying drawings, or applying them directly or indirectly in other related technical fields, are similarly included in the scope of patent protection of the present disclosure.

What is claimed is:

1. An earphone, comprising:

a shell assembly, wherein the shell assembly is provided with a first accommodation cavity and a second accommodation cavity isolated from each other,

a bone-conducting loudspeaker, accommodated in the first accommodation cavity,

an air-conducting loudspeaker, accommodated in the second accommodation cavity, wherein

the shell assembly is provided with a pressure relief hole for connecting the second accommodation cavity to the external environment, and

the pressure relief hole is configured to extend towards a side where the first accommodation cavity is located.

2. The earphone of claim 1, wherein a projection of the pressure relief hole along a direction perpendicular to an arrangement direction of the first accommodation cavity and the second accommodation cavity overlaps at least partially with the first accommodation cavity.

3. The earphone of claim 1, wherein the pressure relief hole is configured to extend inclinedly towards the side where the first accommodation cavity is located, and

the pressure relief has a dimension along a direction perpendicular to an arrangement direction of the first accommodation cavity and the second accommodation cavity, the dimension of the pressure relief hole gradually increases along a direction from the second accommodation cavity to the external environment.

4. The earphone of claim 1, wherein the shell assembly includes a partition wall for isolating the first accommodation cavity and the second accommodation cavity, and at least a portion of the partition wall extends toward the side where the first accommodation cavity is located and forms a portion of a wall of the pressure relief hole.

5. The earphone of claim 4, wherein on a cross-section perpendicular to a vibration direction of the bone-conducting loudspeaker, the bone-conducting loudspeaker includes a long-side edge and a short-side edge, wherein

a dimension of the long-side edge is larger than a dimension of the short-side edge.

6. The earphone of claim 5, wherein the partition wall includes a first wall segment and a second wall segment,

the first wall segment and the long-side edge are arranged side by side,

the second wall segment is connected to the first wall segment and extends inclinedly from a periphery of a junction between the long-side edge and the short-side edge towards the side where the first accommodation cavity is located, and

the second wall segment is configured to form the portion of the wall of the pressure relief hole.

7. The earphone of claim 4, wherein the shell assembly includes a first shell, a second shell, and a third shell, the partition wall is disposed on the first shell and/or the second shell,

the first shell and the second shell fit with each other to form the first accommodation cavity and the pressure relief hole,

the first shell and/or the second shell further form a portion of the second accommodation cavity, and

the third shell and the first shell and/or the second shell fit with each other to form another portion of the second accommodation cavity.

8. The earphone of claim 7, wherein a vibration direction of the bone-conducting loudspeaker intersects with a vibration direction of the air-conducting loudspeaker, the first shell and the second shell fit with each other along the vibration direction of the bone-conducting loudspeaker, and the third shell and the first shell and/or the second shell fit with each other along the vibration direction of vibration of the air-conducting loudspeaker.

9. The earphone of claim 8, wherein the partition wall is disposed on the first shell, wherein

the first shell is provided with a first sub-accommodation cavity and a second sub-accommodation cavity arranged on opposite sides of the partition wall,

an opening direction of the first sub-accommodation cavity is disposed along a wall surface of the partition wall, and

an opening direction of the second sub-accommodation cavity intersects with the wall surface of the partition wall.

10. The earphone of claim 9, wherein

the second shell is provided with a third sub-accommodation cavity,

the second shell is capped at an open end of the first sub-accommodation cavity, and

the third sub-accommodation cavity fits with the first sub-accommodation cavity to form the first accommodation cavity.

11. The earphone of claim 9, wherein

the third shell is provided with a fourth sub-accommodation cavity,

the third shell is capped at an open end of the second sub-accommodation cavity, and

the fourth sub-accommodation cavity fits with the second sub-accommodation cavity to form the second accommodation cavity.

12. The earphone of claim 4, wherein the air-conducting loudspeaker includes the vibration diaphragm,

the vibration diaphragm separates the second accommodation cavity into a first cavity portion disposed on a side of the vibration diaphragm away from the partition wall and a second cavity portion disposed between the vibration diaphragm and the partition wall,

the pressure relief hole is connected to the second cavity portion, and

the shell assembly is further provided with a sound outlet hole connected to the first cavity portion.

13. The earphone of claim 12, wherein a distance between the vibration diaphragm and the partition wall along a vibration direction of the air-conducting loudspeaker is greater than 0.8 mm.

14. The earphone of claim 12, wherein the air-conducting loudspeaker further includes the magnetic circuit assembly and the voice coil,

the magnetic circuit assembly includes the shield with the open end, and the annular flange that is disposed at the open end of the shield and protrudes from the outer periphery surface of the shield,

the rim of the vibration diaphragm is fixed to a side of the annular flange towards the open end and the vibration diaphragm covers the open end of the shield,

the voice coil is connected to the vibration diaphragm, and the vibration diaphragm is disposed away from the sound outlet hole compared to the shield.

15. The earphone of claim 12, wherein a center of the pressure relief hole is not higher than a center of the sound outlet hole, and a line connecting the center of the pressure relief hole and the center of the sound outlet hole points towards the user's ear in a wearing state.

16. The earphone of claim 12, wherein the shell assembly includes a plurality of sound outlet holes, a line connecting a center of the pressure relief hole and a center of the plurality of sound outlet holes points towards the user's ear in a wearing state, the center of the plurality of sound outlet holes refers to a center of a shape formed by lines connecting a center of every sound outlet hole of the plurality of sound outlet holes.

17. The earphone of claim 12, wherein in the wearing state, an angle between the line connecting the center of the pressure relief hole and the center of the sound outlet hole and a coronal axis of the user is $\geq 0^\circ$ and $< 90^\circ$, and an angle between the line connecting the center of the pressure relief hole and the center of the sound outlet hole and a vertical axis of the user is $> 0^\circ$ and $\leq 90^\circ$.

18. The earphone of claim 1, wherein a central axis of the bone-conducting loudspeaker and a central axis of the air-conducting loudspeaker are skew lines, and along a direction perpendicular to the central axis of the bone-conducting loudspeaker and the central axis of the air-conducting loudspeaker, the central axis of the bone-conducting speaker is located farther from the pressure relief hole compared to the central axis of the air-conducting loudspeaker.

19. The earphone of claim 18, wherein a distance between the central axis of the bone-conducting loudspeaker and the central axis of the air-conducting loudspeaker is in a range of 0.2 mm to 1.2 mm.

20. The earphone of claim 1, wherein a sealing performance of the first accommodation cavity is greater than a sealing performance of the second accommodation cavity.

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